

Gottfried Wilhelm Leibniz

1646-1716

Self-Study Notes

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For every reader learning to make mathematics their own.

Contents

1	Differentiation	1
1.1	The Tangent Problem	1
1.2	The Derivative	3
1.2.1	First Consequences	9
1.3	One-Sided Derivatives	12
1.4	Carathéodory and the Chain Rule	15
1.4.1	Carathéodory Characterization	15
1.5	The Shape Questions	20
1.5.1	Interior Extrema	21
1.6	The Mean Value Theorems	25
1.7	Reading the Graph from f' and f''	31
1.7.1	First-Order Shape	31
1.7.2	Second-Order Shape	37
1.7.3	Higher-Order Derivatives	37
1.7.4	Second-Order Shape	38
1.7.5	Darboux's Theorem	41
1.7.6	Smoothness Classes	41
1.8	The Algebra of Derivatives	49
1.8.1	From Characterization to Computation	49
1.8.2	Closure	52
1.8.3	Global Forms on an Interval	54
1.8.4	Inverting the Operation	58
1.8.5	Indeterminate Forms: L'Hôpital's Rule	60
1.9	Taylor: Construction and Its Error	61
1.9.1	The Differential	66

Chapter 1

Differentiation



1.1 The Tangent Problem

Secant and Tangent — Quick Reference

Concept	Meaning
Secant line	Line through two points of a graph
Tangent line	Limiting local line determined by the derivative

Remark (Section guide). Before the derivative is a formula it is a question: what straight line best matches a curve at a single point? The route in is indirect: choose a second point, draw the computable secant line, and let that point slide toward the first. If the secants settle toward a limiting line, that line is the tangent and its slope is the derivative.

Definition (Secant Line)

Definition 1.1.1 (Secant Line). Let $f : A \rightarrow \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A (x_1 \neq x_2 \Rightarrow \text{SecantLine}(f, x_1, x_2) \text{ has slope } \frac{f(x_2) - f(x_1)}{x_2 - x_1}).$$

Remark (Interpretation). With increments $\Delta x := x_2 - x_1$ and $\Delta y := f(x_2) - f(x_1)$, the secant slope is $\Delta y / \Delta x$ — the average rate of change of f over the interval $[x_1, x_2]$. This is an

exact algebraic identity; no limit is involved. As $x_2 \rightarrow x_1$, if the secant slope converges, its limit is the derivative $f'(x_1)$ — the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point $(x_1, f(x_1))$ as x_2 approaches x_1 , and the limit position is the tangent line.

Definition (Difference Quotient)

Definition 1.1.2 (Difference Quotient). Let $f : A \rightarrow \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of f from c to x is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

Remark (Interpretation). The difference quotient is the slope of the secant line through $(c, f(c))$ and $(x, f(x))$. It is the quantity whose limiting value, when it exists as $x \rightarrow c$, becomes the derivative.

Remark (Dependencies).

- [Secant Line](#)

Definition (Tangent Line)

Definition 1.1.3 (Tangent Line). Let $f : A \rightarrow \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c . The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

Remark (Interpretation). The tangent line is the limit of the family of secant lines through $(c, f(c))$ and $(x, f(x))$ as $x \rightarrow c$. More precisely: the slope $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$ converges to $f'(c)$, so the tangent line is the line whose slope is that limit. It is also the unique linear function $\ell(x) = f(c) + f'(c)(x - c)$ satisfying $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$ — the best first-order approximation to f near c . The error $f(x) - \ell(x) = o(x - c)$ as $x \rightarrow c$, meaning it vanishes faster than the displacement.

Remark (Dependencies).

- [Definition: Secant Line](#)
- [Definition: Derivative at a Point](#)

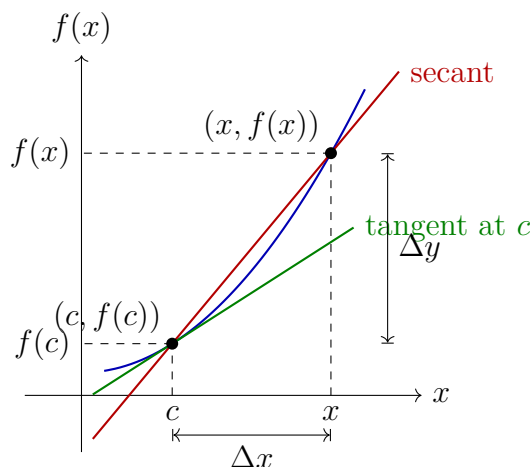


Figure 1.1: The secant line through $(c, f(c))$ and $(x, f(x))$ has slope $\Delta y / \Delta x$. As $x \rightarrow c$, the secant line rotates toward the tangent line at $(c, f(c))$, whose slope is $f'(c)$.

1.2 The Derivative

Derivative Definition — Quick Reference

Concept	Meaning
Derivative at a point	Epsilon–delta limiting secant slope
Example: derivative of x^2	The definition in a removable-singularity computation
Topological derivative	Neighbourhood formulation of the same limit
Sequential derivative	Sequence test for differentiability
Equivalent definitions	Epsilon–delta, topological, and sequential forms agree
h -form	Difference quotient written with $x = c + h$
Differentiability implies continuity	Differentiability is stronger than continuity
Example: one corner	Continuous but not differentiable at one point
Example: sawtooth	Continuous with infinitely many isolated corners
Example: Weierstrass function	Continuous and nowhere differentiable
Uniqueness	The derivative value is single-valued
Identity derivative	Base case $D(x) = 1$
Constant derivative	Base case $D(C) = 0$

Remark (Section guide). The derivative is the number the secant slopes approach, but the word “limit” has three useful faces. The epsilon–delta form is the contract; the topological form strips it to neighbourhoods; the sequential form is the disproving tool, since one bad approach sequence is enough.

Definition (Derivative at a Point)

Definition 1.2.1 (Derivative at a Point). Let $f : D \rightarrow \mathbb{R}$ be a real-valued function and $c \in \text{int}(D)$. We say f is *differentiable at c* with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D \left(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D \left(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

Remark (Definition predicate reading).

$$\text{DerivativeAt}(f, c, L) \iff \text{FunctionLimit} \left(\lambda x. \frac{f(x) - f(c)}{x - c}, D, c, L \right).$$

The predicate $\text{DerivativeAt}(f, c, L)$ asserts that L is the value of the limit of the difference quotient of f at c .

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall \delta > 0 \exists x \in D \left(0 < |x - c| < \delta \wedge \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

Remark (Negation predicate reading).

$$\neg \text{DerivativeAt}(f, c, L) \iff \neg \text{FunctionLimit} \left(\lambda x. \frac{f(x) - f(c)}{x - c}, D, c, L \right).$$

Remark (Failure modes). • The difference quotient has no finite limit at c (e.g. $|f(x) - f(c)|/|x - c| \rightarrow \infty$).

- The difference quotient oscillates without settling near any single value.
- The proposed value L is the wrong candidate.

Remark (Interpretation). The derivative is a limit of slopes of secant lines. The ε - δ form demands that the difference quotient can be made arbitrarily close to L by taking x sufficiently close to c . The input condition $0 < |x - c|$ is punctured: $x = c$ is excluded because $(f(c) - f(c))/(c - c)$ is $0/0$.

Remark (Dependencies).

- [Limit of a Function](#)

Example (Derivative of x^2 from the Definition)

Remark (Worked example). Let $f(x) = x^2$ and fix $c \in \mathbb{R}$. For $x \neq c$,

$$\frac{f(x) - f(c)}{x - c} = \frac{x^2 - c^2}{x - c} = x + c.$$

The singularity has been removed by exact algebra, so the limit as $x \rightarrow c$ is $2c$. Hence $f'(c) = 2c$. Read through Carathéodory, this is the factor $\varphi(x) = x + c$, continuous at c with $\varphi(c) = 2c$.

Definition (Topological Definition of Derivative at a Point)

Definition 1.2.2 (Topological Definition of Derivative at a Point). Let $f : D \rightarrow \mathbb{R}$ and $c \in \text{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L , there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Standard quantified statement).

$$\forall V \text{ neighbourhood of } L \exists U \text{ neighbourhood of } c \forall x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Definition predicate reading).

$$\text{DerivativeAt}_{\text{top}}(f, c, L) \iff \forall V \ni L \exists U \ni c \forall x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L \forall U \text{ neighbourhood of } c \exists x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Negation predicate reading).

$$\neg \text{DerivativeAt}_{\text{top}}(f, c, L) \iff \exists V \ni L \forall U \ni c \exists x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Failure modes). The topological derivative fails when a fixed output neighbourhood of L cannot be entered by the difference quotient no matter how tightly the input is restricted around c . Geometric reading: no punctured neighbourhood around c maps all secant slopes into any prescribed window around L .

Remark (Interpretation). The topological form replaces ε -balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

Remark (Dependencies).

- Function
- [Relative Neighborhood](#)
- Subtraction on $\mathbb{R}$
- Division on $\mathbb{R}$

Definition (Sequential Definition of Derivative at a Point)

Definition 1.2.3 (Sequential Definition of Derivative at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . We say that f is differentiable at c with derivative L if

$$\forall(x_n) \subseteq A \setminus \{c\} \left(x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L \right).$$

Remark (Standard quantified statement).

$$\forall(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L.$$

Remark (Definition predicate reading).

$$\text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \forall(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L.$$

Remark (Negated quantified statement).

$$\exists(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L.$$

Remark (Negation predicate reading).

$$\neg \text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \exists(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L.$$

Remark (Failure modes). Failure occurs when a single approach sequence $(x_n) \rightarrow c$ exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than L (wrong slope candidate). A single such sequence certifies non-differentiability.

Remark (Failure mode decomposition).

$$\underbrace{x_n \rightarrow c}_{\text{approach condition}} \wedge \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g., $f(x) = |x|$ at $c = 0$), or quotients growing without bound.

Remark (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence $x_n \rightarrow c$ along which the difference quotient fails to converge to any single value.

Remark (Dependencies).

- [Sequence](#)
- [Convergent Sequence](#)
- [Derivative at a Point](#)
- [Limit of a Function](#)

Theorem (Equivalence of Definitions of the Derivative at a Point)

Theorem 1.2.4 (Equivalence of Definitions of the Derivative at a Point). *Let $f : A \rightarrow \mathbb{R}$ and $c \in A$ be a limit point of A . The following are equivalent:*

1. *f is differentiable at c in the ε - δ sense with derivative L .*
2. *f is differentiable at c in the topological sense with derivative L .*
3. *f is differentiable at c in the sequential sense with derivative L .*

Go to proof.

Remark (Standard quantified statement).

$$(i) \iff (ii) \iff (iii).$$

Remark (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \iff \text{DerivativeAt}_{\text{top}}(f, c, L) \iff \text{DerivativeAt}_{\text{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of f at c .

Remark (Negated quantified statement).

$$\begin{aligned} \neg(i) &\iff \neg(ii) \iff \neg(iii) \\ &\iff \exists(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L. \end{aligned}$$

Remark (Failure modes). The equivalence cannot produce contradictory verdicts: if c is a limit point of A , all three forms agree. Degenerate case: if c is not a limit point of A , the punctured-ball and sequence conditions are vacuously satisfied by every candidate L simultaneously, making the equivalence trivially true for all L rather than characterizing a unique derivative.

Remark (Interpretation). The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

Remark (Dependencies).

- [Derivative at a Point](#)
- [Topological Definition](#)
- [Sequential Definition](#)

Proposition (Derivative: h -Form Equivalence)

Proposition 1.2.5 (Derivative: h -Form Equivalence). *Let $f : A \rightarrow \mathbb{R}$ and $c \in \text{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \iff \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = L.$$

Remark (Negated quantified statement).

$$\neg \text{DerivativeAt}(f, c, L) \iff \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \neq L.$$

Remark (Failure modes). The h -form fails in two branches: (i) The incremental ratio $(f(c+h) - f(c))/h$ has no finite limit as $h \rightarrow 0$ (corner, cusp, or vertical tangent at c). (ii) The ratio converges to a finite value different from L (proposed L is the wrong slope).

Remark (Interpretation). The h -form substitutes $x = c + h$ in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

Remark (Dependencies).

- [Derivative at a Point](#)

Remark (Discrete and continuous change). The forward difference operator

$$\Delta f(n) = f(n+1) - f(n)$$

is the discrete analogue of the derivative. The h -form above is its continuous descendant: replace the fixed step 1 by a shrinking step h , divide by h , and pass to the limit.

The kinship matters, but so does the divergence. The discrete product rule has an unavoidable shift:

$$\Delta(fg)(n) = (\Delta f)(n)g(n) + f(n+1)(\Delta g)(n).$$

The continuous product rule loses this shift because $f(x+h) \rightarrow f(x)$ as $h \rightarrow 0$. Likewise, ordinary powers are not the clean basis for Δ ; falling factorials are:

$$\Delta x^{\underline{k}} = k x^{\underline{k-1}}.$$

Passing to the limit is therefore not just a sharpening of a quotient. It restores symmetries that the discrete operator must carry as shifts.

1.2.1 First Consequences

Theorem (Differentiable Implies Continuous)

Theorem 1.2.6 (Differentiable Implies Continuous). *Let $f : A \rightarrow \mathbb{R}$ and $c \in A$. If f is differentiable at c , then f is continuous at c .*

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivableAt}(f, c) \Rightarrow \text{ContinuousAt}(f, c).$$

Remark (Predicate reading).

$$\text{DifferentiableAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

Remark (Failure modes). The implication can only fail in the reverse direction: continuity at c does not force differentiability at c .

Remark (Failure mode decomposition). The converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable there.

Remark (Contrapositive quantified statement).

$$\neg \text{ContinuousAt}(f, c) \Rightarrow \neg \text{DerivableAt}(f, c).$$

Remark (Contrapositive predicate reading).

$$\neg \text{ContinuousAt}(f, c) \implies \neg \text{DifferentiableAt}(f, c).$$

Remark (Interpretation). Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at $f(c)$ as $x \rightarrow c$.

Remark (Dependencies).

- [Derivative at a Point](#)
- [Continuity at a Point](#)

Remark (Converse failure). The theorem runs one way only. Differentiability forces continuity, but continuity does not force differentiability. The negation of the converse is

$$\exists f \exists c : \text{ContinuousAt}(f, c) \wedge \neg \text{DerivableAt}(f, c).$$

The examples below escalate the failure: one corner, infinitely many isolated corners, and no differentiability anywhere.

Example (One Corner: $|x|$)

Remark (Worked example). The function $f(x) = |x|$ is continuous on $\mathbb{R}$, but it is not differentiable at 0. Its one-sided derivatives are -1 and $+1$, so the two one-sided slopes do not agree. Thus a single corner is enough to separate continuity from differentiability.

Example (Finitely Many Corners: A Sawtooth)

Remark (Worked example). On a compact interval, a triangular sawtooth made from finitely many line segments can be continuous while failing to be differentiable at each corner. For example,

$$s(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2 - x, & 1 \leq x \leq 2, \end{cases}$$

is continuous on $[0, 2]$, but the left derivative at 1 is 1 and the right derivative is -1 . Adding more teeth gives any finite number of corner failures without breaking continuity.

Example (The Weierstrass Function)

Remark (Worked example). For $0 < a < 1$ and an odd integer b satisfying $ab > 1 + \frac{3}{2}\pi$, the Weierstrass function

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

is continuous on $\mathbb{R}$, because the series converges uniformly by the Weierstrass M -test. Classically, under the displayed condition, it is differentiable at no point. Thus continuity may hold everywhere while differentiability holds nowhere.

Remark (Proof status). The continuity of W follows from uniform convergence once the function-series machinery is available. The nowhere-differentiability proof is a classical result of Weierstrass and is deferred to the function-series development; it is used here only as a counterexample to the converse of Differentiable Implies Continuous.

Remark (Forward flag). Brownian motion has sample paths that are almost surely continuous everywhere and differentiable nowhere. This is the stochastic analogue of the Weierstrass phenomenon and one reason ordinary calculus must be replaced by the Itô integral in stochastic analysis.

Theorem (Uniqueness of the Derivative)

Theorem 1.2.7 (Uniqueness of the Derivative). *Let $f : A \rightarrow \mathbb{R}$ and $c \in A$. If f is differentiable at c , its derivative is unique.*

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge \text{DerivativeAt}(f, c, L') \Rightarrow L = L'.$$

Remark (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \wedge \text{DerivativeAt}(f, c, L') \implies L = L'.$$

Remark (Failure modes). The uniqueness proof assumes $L \neq L'$ and takes $\varepsilon = |L - L'|/2 > 0$. The definition supplies δ making the difference quotient within ε of both L and L'

simultaneously, forcing $|L - L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster-point hypothesis on c is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

Remark (Contrapositive quantified statement).

$$L \neq L' \Rightarrow \neg(\text{DerivativeAt}(f, c, L) \wedge \text{DerivativeAt}(f, c, L')).$$

Two distinct candidates cannot both be the derivative of f at c .

Remark (Contrapositive predicate reading).

$$L \neq L' \implies \neg \text{DerivativeAt}(f, c, L) \vee \neg \text{DerivativeAt}(f, c, L').$$

Remark (Interpretation). Uniqueness licenses writing $f'(c)$ as a definite value rather than a set of candidates.

Remark (Dependencies).

- [Derivative at a Point](#)

Proposition (Derivative of the Identity)

Proposition 1.2.8 (Derivative of the Identity). *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall c \in \mathbb{R} : \frac{d}{dx}[x] \Big|_{x=c} = 1.$$

Remark (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

Remark (Dependencies).

- [Derivative at a Point](#)
- [Function](#)

Proposition (Derivative of a Constant)

Proposition 1.2.9 (Derivative of a Constant). *Let $k \in \mathbb{R}$ and let $f(x) = k$ for all $x \in \mathbb{R}$. Then $f'(c) = 0$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall k \in \mathbb{R} \forall c \in \mathbb{R} : \frac{d}{dx}[k] \Big|_{x=c} = 0.$$

Remark (Interpretation). A constant function has zero change across every secant line, so its instantaneous rate of change is zero everywhere. Together with the derivative of the identity, this is one of the base cases for the algebraic computation rules.

Remark (Dependencies).

- [Derivative at a Point](#)

1.3 One-Sided Derivatives

One-Sided Derivatives — Quick Reference

Concept	Meaning
Left-hand derivative	Limiting slope approached from the left
Right-hand derivative	Limiting slope approached from the right
Two-sided criterion	Differentiability iff both one-sided derivatives agree
Example: x at 0	A continuous corner with unequal one-sided slopes

Remark (Section guide). A corner is a place where a curve has a left answer and a right answer and refuses to choose. One-sided derivatives interrogate each side alone, and their agreement criterion is the fastest way to certify many failures of differentiability.

Definition (Left-Hand Derivative)

Definition 1.3.1 (Left-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I . We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R} \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I \left(-(\delta) < x - c < 0 \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

In this case $f'_-(c) = L$.

Remark (Definition predicate reading).

$$\text{LeftDerivativeAt}(f, c, L) \iff \text{LeftLimit} \left(\lambda x. \frac{f(x) - f(c)}{x - c}, c, L \right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall \delta > 0 \exists x \in I \left(-\delta < x - c < 0 \wedge \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R} \neg \text{LeftDerivativeAt}(f, c, L).$$

No finite L is the left-hand limit of the difference quotient.

Remark (Interpretation). The left-hand derivative $f'_-(c)$ is the limit of the difference quotient as x approaches c *strictly from the left*. It measures the instantaneous rate of change of f at c as seen from the left side. The right-hand derivative $f'_+(c)$ is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative $f'(c)$ exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for $f(x) = |x|$ at $c = 0$, $f'_-(0) = -1$ and $f'_+(0) = +1$; since these disagree, $f'(0)$ does not exist.

Remark (Dependencies).

- **Definition: Derivative at a Point**

Definition (Right-Hand Derivative)

Definition 1.3.2 (Right-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I . We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R} \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I \left(0 < x - c < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

In this case $f'_+(c) = L$.

Remark (Definition predicate reading).

$$\text{RightDerivativeAt}(f, c, L) \iff \text{RightLimit} \left(\lambda x. \frac{f(x) - f(c)}{x - c}, c, L \right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall \delta > 0 \exists x \in I \left(0 < x - c < \delta \wedge \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R} \neg \text{RightDerivativeAt}(f, c, L).$$

Remark (Interpretation). The right-hand derivative $f'_+(c)$ is the instantaneous rate of change seen from inputs larger than c . It is the endpoint derivative available at the left endpoint of a closed interval, and it is one half of the two-sided derivative test at interior points.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Definition: Left-Hand Derivative](#)

Theorem

Theorem 1.3.3 (Differentiability and One-Sided Derivatives). *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \left(\text{LeftDerivativeAt}(f, c, L) \wedge \text{RightDerivativeAt}(f, c, L) \right) \text{ for some } L \in \mathbb{R}.$$

Remark (Negated quantified statement).

$$\neg \text{DifferentiableAt}(f, c) \iff \neg \left(\text{LeftDerivativeAt}(f, c, L) \wedge \text{RightDerivativeAt}(f, c, L) \right) \text{ for all } L \in \mathbb{R}.$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

Remark (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show f is not differentiable at c , compute $f'_-(c)$ and $f'_+(c)$ and observe they disagree. For $f(x) = |x|$ at $c = 0$: $f'_-(0) = -1 \neq +1 = f'_+(0)$, so f is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval $[a, b]$: differentiable on the open interior (a, b) in the full sense, plus $f'_+(a)$ and $f'_-(b)$ both exist at the endpoints.

Remark (Dependencies).

- [Definition: Left-Hand Derivative](#)
- [Definition: Right-Hand Derivative](#)
- [Definition: Derivative at a Point](#)

Example ($|x|$ Has No Derivative at 0)

Remark (Worked example). For $f(x) = |x|$ at $c = 0$, the one-sided difference quotients are

$$\frac{|x| - 0}{x - 0} = \frac{|x|}{x} = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

Thus $f'_-(0) = -1$ and $f'_+(0) = 1$. The two values disagree, so the two-sided derivative does not exist at 0, even though f is continuous there.

1.4 Carathéodory and the Chain Rule

Carathéodory and Chain Rule — Quick Reference

Concept	Role
Carathéodory factorization	Continuous slope factor for differentiability
Carathéodory characterization	Differentiability iff the slope factor extends continuously
Chain Rule	Derivative of a composition
Leibniz Rule	n -th derivative of a product
Faa di Bruno	n -th derivative of a composition
Faa di Bruno at $n = 2$	Coefficient check for the second derivative
Faa di Bruno at $n = 3$	Coefficient check for the third derivative

1.4.1 Carathéodory Characterization

Remark (Section guide). Carathéodory replaces division by factorisation: $f(x) - f(c) = (x - c)\varphi(x)$, with φ continuous at c . Differentiability becomes continuity of the slope factor, and continuity composes. That is why the chain rule becomes clean rather than a fight with vanishing denominators.

Definition (Carathéodory Factorization)

Definition 1.4.1 (Carathéodory Factorization). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \rightarrow \mathbb{R}$, continuous at c , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Remark (Standard quantified statement).

$$\exists \phi : I \rightarrow \mathbb{R} \left(\text{ContinuousAt}(\phi, c) \wedge \forall x \in I [f(x) = f(c) + (x - c)\phi(x)] \right).$$

Remark (Definition predicate reading). The existence of a Carathéodory factorization at c is equivalent to $\text{DifferentiableAt}(f, c)$ by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \frac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of f at c is precisely the statement that this piecewise-defined ϕ is continuous at c .

Remark (Interpretation). The Carathéodory factorization replaces the difference quotient — a function with a removable discontinuity at c — with a function ϕ that is genuinely continuous at c . The factorization $f(x) - f(c) = (x - c)\phi(x)$ holds everywhere on I , including at $x = c$.

where both sides equal zero. The derivative is recovered as $\phi(c)$ without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

Carathéodory vs. Lipschitz. Both control the increment $f(x) - f(c)$ by the factor $(x - c)$, but differently. The Carathéodory factorization requires the slope function $\phi(x)$ to extend *continuously* to c ; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition $|f(x) - f(c)| \leq L|x - c|$ only bounds the size of the increment. Carathéodory implies Lipschitz: if ϕ is continuous at c it is bounded near c , so $|f(x) - f(c)| = |\phi(x)||x - c| \leq K|x - c|$ for some K near c . The converse fails: $f(x) = |x|$ is Lipschitz on $\mathbb{R}$ with constant 1 but the slope function takes values ± 1 on either side of 0 with no continuous extension.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Definition: Continuous at a Point](#)

Theorem

Theorem 1.4.2 (Carathéodory Characterization of Differentiability). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. The following are equivalent:*

- (i) *f is differentiable at c .*
- (ii) *There exists a function $\phi : I \rightarrow \mathbb{R}$, continuous at c , such that*

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} \text{DifferentiableAt}(f, c) &\iff \exists \phi : I \rightarrow \mathbb{R} \left(\text{ContinuousAt}(\phi, c) \right. \\ &\quad \left. \wedge \forall x \in I [f(x) = f(c) + (x - c)\phi(x)] \right). \end{aligned}$$

In this case $\phi(c) = f'(c)$.

Remark (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define ϕ as the difference quotient extended by $f'(c)$ at c and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from ϕ and read off its limit as $\phi(c)$. The theorem's power is that it converts every future differentiability argument into a continuity argument: to show $g \circ f$ is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of f and g yields one automatically.

Remark (Dependencies).

- [Definition: Carathéodory Factorization](#)
- [Definition: Derivative at a Point](#)
- [Definition: Continuous at a Point](#)

Theorem

Theorem 1.4.3 (Chain Rule). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \rightarrow \mathbb{R}$ and $g : J \rightarrow \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c , and*

$$(g \circ f)'(c) = g'(f(c)) f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, f(c)) \implies (g \circ f)'(c) = g'(f(c)) f'(c).$$

Remark (Interpretation). The derivative of a composition is the derivative of the outside function evaluated at the inside value, multiplied by the derivative of the inside function.

Remark (Interpretation). Let ϕ_f and ϕ_g be the Carathéodory slope factors for f at c and g at $f(c)$. Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x))(f(x) - f(c)) = \phi_g(f(x))\phi_f(x)(x - c).$$

The new slope factor is

$$\phi_{g \circ f}(x) = \phi_g(f(x))\phi_f(x),$$

which is continuous at c and has value $g'(f(c))f'(c)$. This proof never divides by $f(x) - f(c)$, so it avoids the failure in the informal $\Delta g / \Delta f \cdot \Delta f / \Delta x$ argument when $f(x) = f(c)$ near c .

Remark (Dependencies).

- [Theorem: Carathéodory Characterization of Differentiability](#)
- [Theorem: Differentiable Implies Continuous](#)
- [Theorem: Composition of Limits](#)
- [Theorem: Product Rule for Function Limits](#)

Theorem (Leibniz Rule)

Theorem 1.4.4 (Leibniz Rule). *Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \rightarrow \mathbb{R}$ are n -times differentiable on a neighbourhood of x . Then fg is n -times differentiable at x , and*

$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Diff}^n(f, x) \wedge \text{Diff}^n(g, x) \implies (fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Remark (Interpretation). The binomial pattern is not an accident. Each of the n differentiations lands on either f or g , and $\binom{n}{k}$ counts the ways to send exactly k differentiations to f . At $n = 1$, this is the ordinary product rule.

Remark (Dependencies).

- [Theorem: Product Rule](#)
- [Definition: Higher Derivatives](#)

Theorem (Faa di Bruno's Formula)

Theorem 1.4.5 (Faa di Bruno's Formula). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \rightarrow J$ is n -times differentiable on a neighbourhood of x , and suppose $f : J \rightarrow \mathbb{R}$ is n -times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is n -times differentiable at x , and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^n m k_m = n} \frac{n!}{\prod_{m=1}^n k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^n (g^{(m)}(x))^{k_m},$$

where $k_m \geq 0$ and $k = \sum_{m=1}^n k_m$. Go to proof.

Remark (Standard quantified statement).

$$\text{Diff}^n(g, x) \wedge \text{Diff}^n(f, g(x)) \implies (f \circ g)^{(n)}(x) = \sum_{\sum m k_m = n} \frac{n!}{\prod_m k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_m (g^{(m)}(x))^{k_m}.$$

Remark (Interpretation). The chain rule gives the first derivative of a composition. Repeated differentiation is deeper: product-rule terms and chain-rule terms interlace, and the book-keeping is governed by integer partitions. Faa di Bruno's formula is the all-orders chain rule.

Remark (Hypotheses). Smoothness is more than this formula needs. The n -th order statement requires n derivatives of g near x and n derivatives of f near $g(x)$. At $n = 1$, the formula is exactly the chain rule.

Remark (Structural reading). The tuple $(k_1, \dots, k_n)$ records an integer partition of n : k_m is the number of parts of size m . The coefficient

$$\frac{n!}{\prod_m k_m! (m!)^{k_m}}$$

counts set partitions with that block-size profile. The outer derivative $f^{(k)}$ is differentiated once per block, while each block of size m contributes one factor $g^{(m)}$.

Remark (Normalization). The formula uses the bare-product normalization: the coefficient contains $(m!)^{k_m}$, and the product contains $(g^{(m)}(x))^{k_m}$. Equivalently, one may put $g^{(m)}(x)/m!$ inside the product and remove $(m!)^{k_m}$ from the denominator. Mixing these two conventions gives the wrong coefficients.

Remark (Cross-volume callback). The constraint $\sum_m m k_m = n$ is precisely the multiplicity form of an integer partition of n . Thus the number of summands is $p(n)$, the partition function from the discrete and number-system material, now appearing as bookkeeping for iterated differentiation.

Remark (Scope). This theorem is enrichment, not a critical dependency for the integration, measure, or manifold arcs. It is included because it is the complete all-orders chain rule and a useful callback to partitions.

Remark (Dependencies).

- [Theorem: Chain Rule](#)
- [Theorem: Leibniz Rule](#)
- [Definition: Higher Derivatives](#)
- [Definition: Class \$C^k\$](#)

Example (Faa di Bruno at $n = 2$)

Remark (Worked example). The partitions of 2 are represented by $(k_1, k_2) = (2, 0)$ and $(0, 1)$. The first gives $f''(g)(g')^2$, and the second gives $f'(g)g''$. Hence

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g''.$$

The coefficient on $f'(g)g''$ is 1, which is the quickest check that the normalization has not been mixed.

Example (Faa di Bruno at $n = 3$)

Remark (Worked example). The partitions of 3 produce the three terms

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The middle coefficient 3 counts the three ways to choose which differentiation contributes the second-derivative block of g .

1.5 The Shape Questions

Shape Questions — Quick Reference

Concept	Meaning
Increasing at a point	Local right-up, left-down behavior
Decreasing at a point	Local right-down, left-up behavior
Relative minimum	Locally smallest function value
Relative maximum	Locally largest function value
Relative extremum	Relative minimum or maximum
Convex function	Graph lies below its chords
Concave function	Graph lies above its chords
Inflection point	Local change in bending behavior

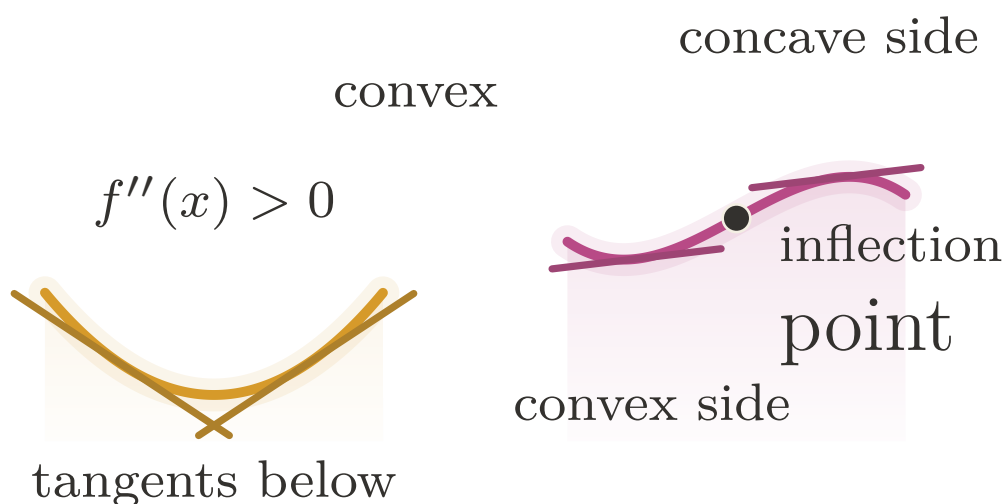


Figure 1.2: Convexity, concavity, and the change of bending at an inflection point.

Definition (Convex Function)

Definition 1.5.1 (Convex Function). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *convex* on I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Remark (Standard quantified statement).

$$\forall x, y \in I \quad \forall \lambda \in [0, 1] : \quad f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Remark (Interpretation). Convexity is defined by chords, not derivatives. The graph lies below the line segment joining any two of its points. Derivative tests are later tools for detecting this geometric condition, not the definition of the condition.

Remark (Dependencies).

- Closed Interval

Definition (Concave Function)

Definition 1.5.2 (Concave Function). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *concave* on I if $-f$ is convex on I .

Remark (Standard quantified statement).

$$\text{ConcaveOn}(f, I) \iff \text{ConvexOn}(-f, I).$$

Remark (Interpretation). Concavity is the downward-bending counterpart of convexity. Equivalently, the graph of a concave function lies above each chord joining two of its points.

Remark (Dependencies).

- Convex Function

Definition (Inflection Point)

Definition 1.5.3 (Inflection Point). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$ be an interior point. The point c is an *inflection point* of f if f changes convexity at c : there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that f is convex on one side of c and concave on the other side.

Remark (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \iff f \text{ changes convexity at } c.$$

Remark (Interpretation). An inflection point is a change in the direction of bending. It is not merely a point where $f''(c) = 0$; that equation is only a necessary condition under additional hypotheses.

Remark (Dependencies).

- Convex Function
- Concave Function

1.5.1 Interior Extrema

Interior Extrema — Quick Reference

Statement	Role
Relative minimum	Local lower comparison point
Relative maximum	Local upper comparison point
Relative extremum	Umbrella term for local optima
Interior Extremum Theorem	Interior differentiable extremum has zero derivative
Necessary condition	Extrema occur at critical behavior
Critical-point corollary	Derivative zero or derivative undefined

Definition (Relative Minimum)

Definition 1.5.4 (Relative Minimum). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \leq f(x) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(c) < f(x) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then f has a *strict relative minimum* at c .

Remark (Standard quantified statement).

$$\exists \delta > 0 \forall x \in A (|x - c| < \delta \implies f(c) \leq f(x)).$$

Remark (Definition predicate reading).

$$\text{LocalMinimum}(f, c) \iff \exists \delta > 0 \forall x \in A (\text{Within}(x, c, \delta) \implies f(c) \leq f(x)).$$

Remark (Interpretation). f has a relative minimum at c if $f(c)$ is the smallest value of f in some open neighbourhood of c within the domain. The word *relative* (or *local*) signals that we are comparing $f(c)$ only to nearby values, not to all values of f globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion — relative maximum — reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

Remark (Dependencies).

- Function
- Subset
- Open Interval
- Order on $\mathbb{R}$

Definition (Relative Maximum)

Definition 1.5.5 (Relative Maximum). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(x) < f(c) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then f has a *strict relative maximum* at c .

Remark (Standard quantified statement).

$$\exists \delta > 0 \forall x \in A (|x - c| < \delta \implies f(x) \leq f(c)).$$

Remark (Definition predicate reading).

$$\text{LocalMaximum}(f, c) \iff \exists \delta > 0 \forall x \in A (\text{Within}(x, c, \delta) \implies f(x) \leq f(c)).$$

Remark (Interpretation). A relative maximum is a point whose function value is at least as large as all nearby function values in the domain.

Remark (Dependencies).

- [Definition: Relative Minimum](#)

Definition (Relative Extremum)

Definition 1.5.6 (Relative Extremum). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c .

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \iff \text{LocalMinimum}(f, c) \vee \text{LocalMaximum}(f, c).$$

Remark (Definition predicate reading).

$$\text{LocalExtremum}(f, c) \iff \text{LocalMinimum}(f, c) \vee \text{LocalMaximum}(f, c).$$

Remark (Interpretation). Relative extrema are the local turning candidates of a graph: nearby values lie all on one side of $f(c)$.

Remark (Dependencies).

- [Definition: Relative Minimum](#)
- [Definition: Relative Maximum](#)

Theorem

Theorem 1.5.7 (Interior Extremum Theorem). Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c , then

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \wedge \text{DifferentiableAt}(f, c) \implies \text{DerivativeAt}(f, c, 0).$$

Remark (Contrapositive quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L \neq 0 \Rightarrow \neg \text{LocalExtremum}(f, c).$$

If $f'(c)$ exists and is nonzero, then c is not a relative extremum.

Remark (Contrapositive predicate reading).

$$\text{DerivativeAt}(f, c, L) \wedge L \neq 0 \Rightarrow \neg \text{LocalMinimum}(f, c) \wedge \neg \text{LocalMaximum}(f, c).$$

Remark (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate — at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for $x > c$ and non-negative for $x < c$; since the derivative is a single limit, it is squeezed to zero. The converse fails: $f'(c) = 0$ does not imply an extremum ($f(x) = x^3$ at $c = 0$). Critical points — where $f'(c) = 0$ or $f'(c)$ does not exist — are candidates for extrema, not guarantees.

Remark (Dependencies).

- [Definition: Relative Extremum](#)
- [Definition: Derivative at a Point](#)
- [Definition: Left-Hand Derivative](#)
- [Definition: Right-Hand Derivative](#)
- [Theorem: Differentiability and One-Sided Derivatives](#)

Theorem

Theorem 1.5.8 (Necessary Condition for an Interior Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c , then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \Rightarrow \neg \text{DifferentiableAt}(f, c) \vee \text{DerivativeAt}(f, c, 0).$$

Remark (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming $f'(c)$ exists, the conclusion is a disjunction: either the derivative fails to exist at c , or it exists and equals zero. This is the correct form for identifying *critical points* — points that are candidates for extrema. The set of critical points is the union of points where f' is zero and points where f' does not exist.

Remark (Dependencies).

- [Theorem: Interior Extremum Theorem](#)

Corollary

Corollary 1.5.9 (Necessary Condition for an Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c , then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \Rightarrow \text{DerivativeAt}(f, c, 0) \vee \neg \text{DifferentiableAt}(f, c).$$

Remark (Interpretation). An extremum can occur only at a critical point: either the derivative vanishes or the derivative fails to exist.

Remark (Dependencies).

- [Theorem: Necessary Condition for an Interior Extremum](#)

1.6 The Mean Value Theorems

Mean Value Theorems — Quick Reference

Statement	Role
Rolle's Theorem	Equal endpoint values force a horizontal tangent
Mean Value Theorem	Some tangent slope equals the secant slope
Cauchy Mean Value Theorem	Compares two functions' rates of change
Derivative bound implies Lipschitz	Turns a slope bound into a global modulus

Remark (Section guide). Rolle's theorem says that a differentiable curve returning to the same height must level off somewhere between. The Mean Value Theorem tilts that picture: between any two points, some interior tangent runs parallel to the chord. Cauchy's version compares two functions at once and later powers L'Hôpital's rule.

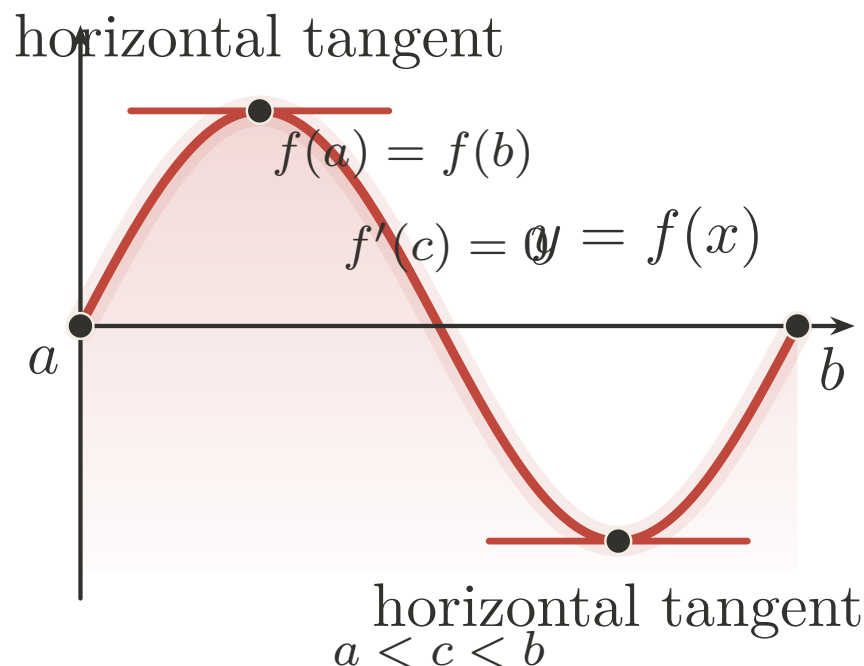


Figure 1.3: Rolle's Theorem: equal endpoint heights force a horizontal tangent.

Theorem (Rolle's Theorem)

Theorem 1.6.1 (Rolle's Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \forall x \in (a, b) \text{ DifferentiableAt}(f, x) \wedge f(a) = f(b) \\ & \Rightarrow \exists c \in (a, b) \text{ DerivativeAt}(f, c, 0). \end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \wedge f(a) = f(b) \\ & \Rightarrow \exists c \in (a, b) : \text{DerivativeAt}(f, c, 0). \end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \text{sgn}(x - 1/2)$ on $[0, 1]$. Then $f(0) = f(1) = -1$, but f has a jump discontinuity at $x = 1/2$. The derivative does not exist anywhere, so there is no horizontal tangent.
- **Differentiability fails:** Let $f(x) = |x - 1/2|$ on $[0, 1]$. Then $f(0) = f(1) = 1/2$, and f is continuous, but f has a corner at $x = 1/2$ where the derivative is undefined. For all $c \neq 1/2$, we have $f'(c) = \pm 1 \neq 0$.

- **Equal endpoints fail:** Let $f(x) = x$ on $[0, 1]$. Then f is continuous and differentiable with $f'(x) = 1 > 0$ everywhere, but $f(0) = 0 \neq 1 = f(1)$.

Remark (Contrapositive quantified statement).

$$(\forall c \in (a, b) : f'(c) \neq 0) \Rightarrow f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b) \vee f(a) \neq f(b).$$

Remark (Contrapositive predicate reading).

$$\begin{aligned} & (\forall c \in (a, b) : \neg \text{DerivativeAt}(f, c, 0)) \\ & \implies \neg \text{ContinuousOn}(f, [a, b]) \vee \neg \text{DifferentiableOn}(f, (a, b)) \vee f(a) \neq f(b). \end{aligned}$$

Remark (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients — the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then f is constant and $f' \equiv 0$ throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

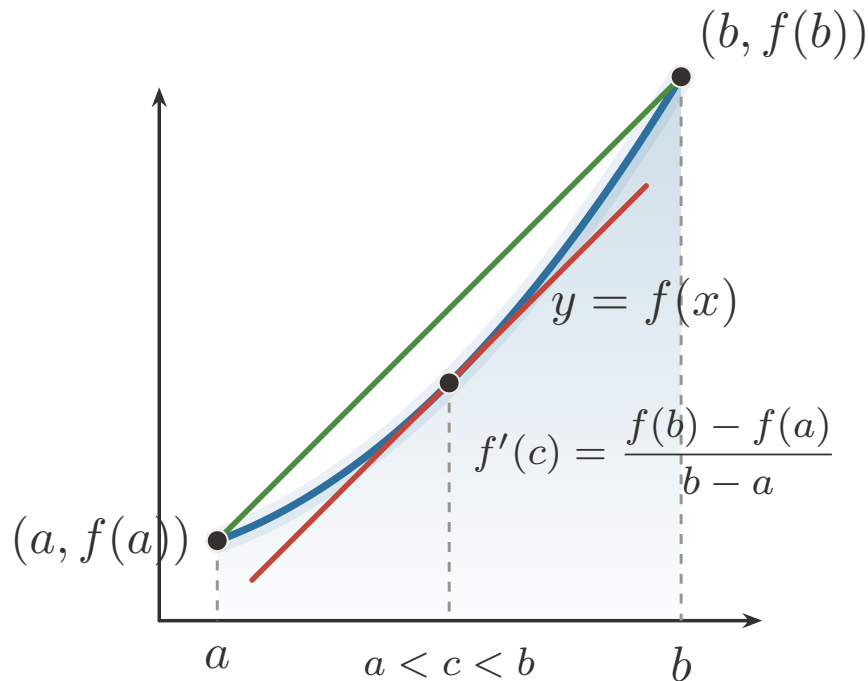


Figure 1.4: Mean Value Theorem: an interior tangent matches the secant slope.

Theorem (Mean Value Theorem)

Theorem 1.6.2 (Mean Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \forall x \in (a, b) \text{ DifferentiableAt}(f, x) \\ & \Rightarrow \exists c \in (a, b) \text{ DerivativeAt}\left(f, c, \frac{f(b)-f(a)}{b-a}\right). \end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \\ & \implies \exists c \in (a, b) : \text{DerivativeAt}\left(f, c, \frac{f(b)-f(a)}{b-a}\right). \end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$. Then f has a jump discontinuity at $x = 1/2$, and the secant slope is $\frac{f(1)-f(0)}{1-0} = 1$. But $f'(x) = 0$ wherever the derivative exists, so no interior point achieves the average rate of change.
- **Differentiability fails:** Let $f(x) = |x|$ on $[-1, 1]$. Then f is continuous and the secant slope is $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$. But $f'(x) = \pm 1$ everywhere f is differentiable (i.e., for $x \neq 0$), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some c exists but gives no formula for it.

Remark (Contrapositive quantified statement).

$$\left(\forall c \in (a, b) : f'(c) \neq \frac{f(b) - f(a)}{b - a} \right) \Rightarrow f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b).$$

Remark (Contrapositive predicate reading).

$$\begin{aligned} & \left(\forall c \in (a, b) : \neg \text{DerivativeAt}\left(f, c, \frac{f(b)-f(a)}{b-a}\right) \right) \\ & \implies \neg \text{ContinuousOn}(f, [a, b]) \vee \neg \text{DifferentiableOn}(f, (a, b)). \end{aligned}$$

Remark (Interpretation). The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point c is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. The proof reduces to Rolle's Theorem by subtracting the secant: define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, which satisfies

$g(a) = g(b) = f(a)$, then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some c exists but gives no formula for it. Its immediate corollaries — the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary — each flow from the MVT by a one-line argument and are the theorem's primary applications.

Remark (Dependencies).

- [Theorem: Rolle's Theorem](#)
- [Definition: Derivative at a Point](#)

Theorem (Cauchy Mean Value Theorem)

Theorem 1.6.3 (Cauchy Mean Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \text{ContinuousOn}(g, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \wedge \text{DifferentiableOn}(g, (a, b)) \\ & \implies \exists c \in (a, b) [(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)]. \end{aligned}$$

Remark (Predicate reading). $\text{CauchyMVT}(f, g, [a, b])$ asserts the existence of an interior point where the velocity vector $(g'(c), f'(c))$ is parallel to the chord vector $(g(b) - g(a), f(b) - f(a))$. Taking $g(x) = x$ recovers the ordinary Mean Value Theorem.

Remark (Negated quantified statement).

$$\forall c \in (a, b) : (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c).$$

Remark (Failure modes). The continuity and differentiability hypotheses fail for the same reasons they fail in the ordinary MVT: jumps break endpoint-to-interior control, and corners leave no derivative to match. The quotient form has the additional nonvanishing hypothesis $g' \neq 0$ on (a, b) ; without it the cross-multiplied conclusion may still hold, but the displayed quotient can be undefined.

Remark (Contrapositive quantified statement).

$$\begin{aligned} & (\forall c \in (a, b) : (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c)) \\ & \implies \neg \text{ContinuousOn}(f, [a, b]) \vee \neg \text{ContinuousOn}(g, [a, b]) \vee \neg \text{DifferentiableOn}(f, (a, b)) \vee \neg \text{DifferentiableOn}(g, (a, b)) \end{aligned}$$

Remark (Contrapositive predicate reading). If no interior point makes the derivative pair $(g'(c), f'(c))$ parallel to the chord pair $(g(b) - g(a), f(b) - f(a))$, then at least one of the four regularity hypotheses fails: continuity of f or g on $[a, b]$, or differentiability of f or g on (a, b) .

Remark (Interpretation). This is the parametric Mean Value Theorem. For the planar curve $t \mapsto (g(t), f(t))$, some interior tangent is parallel to the chord joining its endpoints. It compares two rates of change rather than comparing one function to the horizontal axis. This is the form that later underlies L'Hopital-type ratio arguments and the Cauchy form of Taylor's remainder.

Remark (Dependencies).

- [Theorem: Rolle's Theorem](#)
- [Definition: Derivative at a Point](#)
- [Theorem: Sum Rule](#)
- [Theorem: Constant Multiple Rule](#)

Corollary (Derivative Bound Implies Lipschitz)

Corollary 1.6.4 (Derivative Bound Implies Lipschitz). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \quad \text{for every } x \in I,$$

then f is Lipschitz on I with constant M :

$$|f(x) - f(y)| \leq M|x - y| \quad \text{for every } x, y \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$(\exists M \geq 0 \forall x \in I : |f'(x)| \leq M) \implies \forall x, y \in I : |f(x) - f(y)| \leq M|x - y|.$$

Remark (Predicate reading).

$$\text{BoundedDerivativeOn}(f, I, M) \implies \text{Lipschitz}(f, I, M).$$

Remark (Failure modes). The derivative bound must hold throughout the interval. A function such as $f(x) = \sqrt{x}$ on $[0, 1]$ is continuous and uniformly continuous, but its derivative is unbounded near 0, so no global Lipschitz constant follows from this test. The interval hypothesis is also essential: across a disconnected gap, the MVT cannot compare the two components.

Remark (Interpretation). This corollary converts a pointwise slope bound into a global modulus of continuity. It is the precise bridge from derivative estimates to the Lipschitz condition: controlling the size of f' controls every secant slope on the interval.

Remark (Dependencies).

- [Theorem: Mean Value Theorem](#)
- [Definition: Lipschitz Condition](#)
- [Definition: Derivative at a Point](#)

1.7 Reading the Graph from f' and f''

1.7.1 First-Order Shape

First-Order Shape — Quick Reference

Statement	Role
Zero derivative	Vanishing derivative forces constant behavior
Equal derivatives	Functions differ by a constant
Nonnegative derivative	Detects nondecreasing behavior
Nonpositive derivative	Detects nonincreasing behavior
Positive derivative	Local increasing behavior
Negative derivative	Local decreasing behavior
First derivative test: $\max$	Sign change detects a local maximum
First derivative test: $\min$	Sign change detects a local minimum

Remark (Section guide). The Mean Value Theorem is now available, so the harvest can begin. Vanishing derivatives force constancy, derivative signs force monotonicity, and second derivatives detect bending when enough regularity is present. Darboux and the smoothness hierarchy show that derivatives carry hidden rigidity even when they are not continuous.

Definition (Increasing at a Point)

Definition 1.7.1 (Increasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f is *increasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \, \forall x \in A \cap (c - \delta, c) \, \forall y \in A \cap (c, c + \delta) \, (f(x) \leq f(c) \leq f(y)).$$

Remark (Interpretation). f is increasing at c if c lies between the values of f on either side in some punctured neighbourhood: f is smaller to the left and larger to the right (weakly). This is a strictly local notion — it says nothing about f outside the δ -neighbourhood of c . A function can be increasing at c without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of c , distinguishing this from the notion of a strict local increase.

Remark (Dependencies).

- [Definition: Strictly Increasing Function \(functions chapter\)](#)

Definition (Decreasing at a Point)

Definition 1.7.2 (Decreasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \, \forall x \in A \cap (c - \delta, c) \, \forall y \in A \cap (c, c + \delta) \, (f(x) \geq f(c) \geq f(y)).$$

Remark (Interpretation). The function is decreasing at c when nearby values to the left sit above $f(c)$ and nearby values to the right sit below it.

Remark (Dependencies).

- **Definition:** Increasing at a Point

Theorem (Zero Derivative Implies Constant)

Theorem 1.7.3 (Zero Derivative Implies Constant). Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \rightarrow \mathbb{R}$ be continuous on I and differentiable on (a, b) . If

$$f'(x) = 0 \quad \text{for all } x \in (a, b),$$

then f is constant on I . Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b) \, \text{DerivativeAt}(f, x, 0) \Rightarrow \exists C \in \mathbb{R} \, \forall x \in I \, (f(x) = C).$$

Remark (Contrapositive quantified statement).

$$\neg(\exists C \in \mathbb{R} \, \forall x \in I \, (f(x) = C)) \Rightarrow \exists x \in (a, b) \, \neg \text{DerivativeAt}(f, x, 0).$$

If f is not constant on I , then f' is not identically zero on (a, b) .

Remark (Contrapositive predicate reading).

$$\neg \text{ConstantOn}(f, I) \Rightarrow \exists x \in (a, b) \, (f'(x) \neq 0).$$

Remark (Interpretation). The MVT is the engine here: if $f'(x) = 0$ everywhere on (a, b) , then for any two points $x_1 < x_2$ in I , the MVT supplies a c with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$, so $f(x_1) = f(x_2)$. Since this holds for all pairs, f is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments — if a solution is not identically the zero solution, its derivative is not identically zero somewhere.

Remark (Dependencies).

- Theorem: Mean Value Theorem

Corollary

Corollary 1.7.4 (Equal Derivatives Imply Difference Is Constant). *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \rightarrow \mathbb{R}$ be continuous on I and differentiable on (a, b) . If*

$$f'(x) = g'(x) \quad \text{for all } x \in (a, b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b) (f'(x) = g'(x)) \Rightarrow \exists C \in \mathbb{R} \forall x \in I (f(x) = g(x) + C).$$

Remark (Contrapositive quantified statement).

$$\forall C \in \mathbb{R} \exists x \in I (f(x) \neq g(x) + C) \Rightarrow \exists x \in (a, b) (f'(x) \neq g'(x)).$$

If $f - g$ is not a constant function, then f' and g' disagree somewhere.

Remark (Contrapositive predicate reading).

$$\neg \text{ConstantOn}(f - g, I) \Rightarrow \exists x \in (a, b) (f'(x) \neq g'(x)).$$

Remark (Interpretation). This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line — apply the previous theorem to $h = f - g$, whose derivative is $(f - g)' = f' - g' = 0$.

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

Theorem (Nondecreasing Iff Nonnegative Derivative)

Theorem 1.7.5 (Nondecreasing Iff Nonnegative Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NondecreasingOn}(f, I) \iff \forall x \in I (f'(x) \geq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I (f'(x) < 0) \Rightarrow \neg \text{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

Remark (Contrapositive predicate reading).

$$\exists x \in I (f'(x) < 0) \Rightarrow \neg \text{NondecreasingOn}(f, I).$$

Remark (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction (f nondecreasing $\Rightarrow f' \geq 0$) follows from the definition: the difference quotient is nonnegative when f is nondecreasing, so the limit is nonnegative. The reverse direction ($f' \geq 0 \Rightarrow$ nondecreasing) is a consequence of the MVT: for $x < y$, the MVT gives $f(y) - f(x) = f'(c)(y - x) \geq 0$.

Note carefully: $f' \geq 0$ characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires $f' > 0$ at all but isolated points. The function $f(x) = x^3$ satisfies $f'(0) = 0$ but is strictly increasing; $f' \geq 0$ with equality only at isolated points is sufficient for strict increase, but $f' > 0$ everywhere is not necessary.

Remark (Dependencies).

- [Theorem: Mean Value Theorem](#)
- [Definition: Derivative at a Point](#)

Theorem

Theorem 1.7.6 (Nonincreasing Iff Nonpositive Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NonincreasingOn}(f, I) \iff \forall x \in I (f'(x) \leq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I (f'(x) > 0) \Rightarrow \neg \text{NonincreasingOn}(f, I).$$

Remark (Contrapositive predicate reading).

$$\exists x \in I (f'(x) > 0) \Rightarrow \neg \text{NonincreasingOn}(f, I).$$

Remark (Interpretation). Nonpositive derivative everywhere is exactly the infinitesimal signature of a function that never increases.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

Lemma

Lemma 1.7.7 (Positive Derivative Implies Locally Increasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c . If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L > 0 \Rightarrow \exists \delta > 0 \forall x \in I \left((c - \delta < x < c \Rightarrow f(x) < f(c)) \wedge (c < x < c + \delta \Rightarrow f(x) > f(c)) \right)$$

Remark (Contrapositive quantified statement).

$$\forall \delta > 0 \exists x \in I \left((c - \delta < x < c \wedge f(x) \geq f(c)) \vee (c < x < c + \delta \wedge f(x) \leq f(c)) \right) \Rightarrow f'(c) \leq 0.$$

Remark (Contrapositive predicate reading).

$$\neg \text{LocallyIncreasingAt}(f, c) \Rightarrow f'(c) \leq 0.$$

Remark (Interpretation). If the instantaneous rate of change is strictly positive at c , the function is genuinely climbing through c : it is strictly below $f(c)$ immediately to the left and strictly above immediately to the right. The proof is a direct ε - δ argument: take $\varepsilon = f'(c)/2 > 0$; the definition of the derivative supplies δ such that the difference quotient is within ε of $f'(c)$, forcing it to be positive, which then forces the correct sign on $f(x) - f(c)$. This lemma is the key ingredient in the proof of the First Derivative Test.

Remark (Dependencies).

- Definition: Derivative at a Point

Lemma

Lemma 1.7.8 (Negative Derivative Implies Locally Decreasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c . If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L < 0 \Rightarrow \exists \delta > 0 \forall x \in I \left((c - \delta < x < c \Rightarrow f(x) > f(c)) \wedge (c < x < c + \delta \Rightarrow f(x) < f(c)) \right)$$

Remark (Contrapositive quantified statement).

$$\neg \text{LocallyDecreasingAt}(f, c) \Rightarrow f'(c) \geq 0.$$

Remark (Contrapositive predicate reading).

$$\neg \text{LocallyDecreasingAt}(f, c) \Rightarrow f'(c) \geq 0.$$

Remark (Interpretation). If the derivative at c is strictly negative, nearby values move downward as the input passes through c from left to right.

Remark (Dependencies).

- [Lemma: Positive Derivative Implies Locally Increasing](#)

Theorem (First Derivative Test: Relative Maximum)

Theorem 1.7.9 (First Derivative Test: Relative Maximum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \rightarrow \mathbb{R}$ be continuous on I , and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b) . If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \leq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative maximum at c . Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & \exists \delta > 0 \left[(c - \delta, c + \delta) \subseteq I \wedge \forall x \in (c - \delta, c) (f'(x) \geq 0) \wedge \forall x \in (c, c + \delta) (f'(x) \leq 0) \right] \\ & \Rightarrow \text{LocalMaximum}(f, c). \end{aligned}$$

Remark (Interpretation). If the function is nondecreasing immediately to the left of c and nonincreasing immediately to the right, then c is a local maximum — the function climbs to c and then falls away. The proof integrates: by the nondecreasing characterisation applied on $(c - \delta, c)$, $f(x) \leq f(c)$ for $x < c$; by the nonincreasing characterisation on $(c, c + \delta)$, $f(x) \leq f(c)$ for $x > c$. The sign condition on f' replaces the Interior Extremum Theorem's converse, which does not hold in general; this is a sufficient condition, not a necessary one. The test is applicable even when $f'(c) = 0$ or $f'(c)$ does not exist.

Remark (Dependencies).

- [Theorem: Nondecreasing Iff Nonnegative Derivative](#)
- [Theorem: Nonincreasing Iff Nonpositive Derivative](#)
- [Definition: Relative Maximum](#)

Theorem

Theorem 1.7.10 (First Derivative Test: Relative Minimum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \rightarrow \mathbb{R}$ be continuous on I , and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b) . If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \geq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative minimum at c . Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} \exists \delta > 0 \left[(c - \delta, c + \delta) \subseteq I \wedge \forall x \in (c - \delta, c) (f'(x) \leq 0) \wedge \forall x \in (c, c + \delta) (f'(x) \geq 0) \right] \\ \Rightarrow \text{LocalMinimum}(f, c). \end{aligned}$$

Remark (Interpretation). If the derivative changes from nonpositive to nonnegative at c , the graph falls into c and then rises away from it, so c is a local minimum.

Remark (Dependencies).

- [Theorem: First Derivative Test: Relative Maximum](#)
- [Definition: Relative Minimum](#)

1.7.2 Second-Order Shape

Second-Order Shape — Quick Reference

Concept	Role
Second derivative	Derivative of the derivative
Higher derivatives	Iterated derivatives $f^{(n)}$
Convexity test	$f'' \geq 0$ detects convexity
Concavity test	$f'' \leq 0$ detects concavity
Second derivative test	Classifies critical points
Inflection condition	Continuous f'' changes sign through zero

1.7.3 Higher-Order Derivatives

Definition (Second Derivative)

Definition 1.7.11 (Second Derivative). Let f be differentiable on a neighbourhood of c . If f' is differentiable at c , the *second derivative* of f at c is

$$f''(c) := (f')'(c).$$

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f', c, f''(c)) \iff f''(c) = (f')'(c).$$

Remark (Interpretation). The first derivative measures instantaneous slope. The second derivative measures how that slope itself changes.

Remark (Dependencies).

- [Derivative at a Point](#)

Definition (Higher Derivatives)

Definition 1.7.12 (Higher Derivatives). Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \geq 2$, if $f^{(n-1)}$ is differentiable at c , define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

Remark (Standard quantified statement).

$$\forall n \geq 2 : \quad f^{(n)}(c) = (f^{(n-1)})'(c)$$

whenever the derivative on the right exists.

Remark (Interpretation). Higher derivatives are obtained by iterating the derivative operator. Their existence is not automatic: each stage requires differentiability of the previous derivative.

Remark (Dependencies).

- [Second Derivative](#)

1.7.4 Second-Order Shape

Theorem (Second-Derivative Convexity Test)

Theorem 1.7.13 (Second-Derivative Convexity Test). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be twice differentiable on I . If

$$f''(x) \geq 0 \quad \text{for all } x \in I,$$

then f is convex on I . Go to proof.

Remark (Standard quantified statement).

$$(\forall x \in I, f''(x) \geq 0) \implies \text{ConvexOn}(f, I).$$

Remark (Interpretation). Nonnegative second derivative means the slopes do not decrease. This is the derivative-based signature of convex bending.

Remark (Dependencies).

- [Second Derivative](#)
- [Convex Function](#)

Theorem (Second-Derivative Concavity Test)

Theorem 1.7.14 (Second-Derivative Concavity Test). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be twice differentiable on I . If

$$f''(x) \leq 0 \quad \text{for all } x \in I,$$

then f is concave on I . Go to proof.

Remark (Standard quantified statement).

$$(\forall x \in I, f''(x) \leq 0) \implies \text{ConcaveOn}(f, I).$$

Remark (Interpretation). Nonpositive second derivative means the slopes do not increase. This is the derivative-based signature of concave bending.

Remark (Interpretation). The second-derivative tests are narrower than the chord definitions of convexity and concavity. The function $|x|$ is convex, but it is not twice differentiable at 0.

Remark (Dependencies).

- [Second Derivative](#)
- [Concave Function](#)

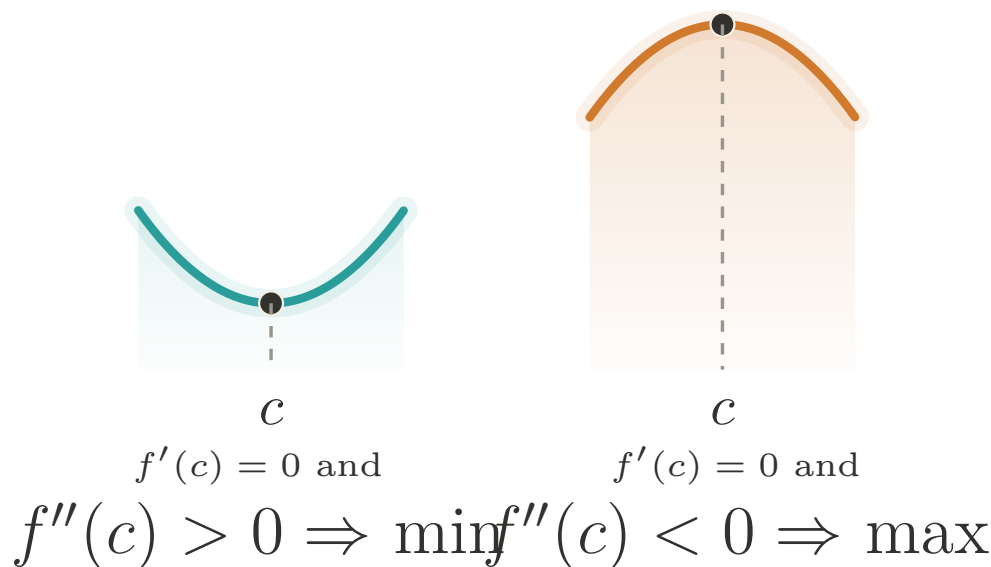


Figure 1.5: Second derivative test: the sign of $f''(c)$ decides the local shape at a critical point.

Theorem (Second Derivative Test)

Theorem 1.7.15 (Second Derivative Test). *Let f be twice differentiable in a neighbourhood of c , and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then f has a strict relative minimum at c .*
- *If $f''(c) < 0$, then f has a strict relative maximum at c .*

Go to proof.

Remark (Standard quantified statement).

$$f'(c) = 0 \wedge f''(c) > 0 \Rightarrow \text{StrictRelativeMinimum}(f, c),$$

and

$$f'(c) = 0 \wedge f''(c) < 0 \Rightarrow \text{StrictRelativeMaximum}(f, c).$$

Remark (Interpretation). At a critical point, positive second derivative means the graph bends upward; negative second derivative means it bends downward.

Remark (Dependencies).

- [Second Derivative](#)
- [Relative Minimum](#)
- [Relative Maximum](#)

Proposition (Inflection Point Necessary Condition)

Proposition 1.7.16 (Inflection Point Necessary Condition). *Let f have an inflection point at c . If f'' exists on a neighbourhood of c and is continuous at c , then*

$$f''(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \wedge \text{ContinuousAt}(f'', c) \implies f''(c) = 0.$$

Remark (Interpretation). An inflection point is a change in bending. When the second derivative is continuous, such a change can pass through c only by passing through zero. The converse can fail: $f''(c) = 0$ does not by itself guarantee an inflection point.

Remark (Dependencies).

- [Inflection Point](#)
- [Second Derivative](#)
- [Continuity at a Point](#)

1.7.5 Darboux's Theorem

Darboux — Quick Reference

Statement	Role
Darboux's Theorem	Derivatives have the intermediate value property

Theorem (Darboux's Theorem)

Theorem 1.7.17 (Darboux's Theorem). *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall k \left(\min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \Rightarrow \exists c \in (a, b) (f'(c) = k) \right).$$

Remark (Interpretation). The Intermediate Value Theorem says continuous functions have the intermediate value property. Darboux's theorem says derivatives also have that property even when the derivative is not continuous. Thus a derivative cannot have a jump discontinuity. The proof uses differentiability of f to construct an auxiliary function $g(x) = f(x) - kx$ and then applies the Extreme Value Theorem and the Interior Extremum Theorem.

Remark (Dependencies).

- [Theorem: Extreme Value Theorem](#)
- [Theorem: Interior Extremum Theorem](#)
- [Definition: Derivative at a Point](#)

1.7.6 Smoothness Classes

Smoothness Classes — Quick Reference

Concept	Meaning
Class C^1	Differentiable with continuous first derivative
Class C^k	k derivatives with continuous top derivative
Class C^∞	Continuous derivatives of every order
Class C^ω	Equal to local Taylor series
Smoothness tower	Strict hierarchy $C^\omega \subsetneq C^\infty \subsetneq \dots$
Class $C^{1,1}$	C^1 with Lipschitz first derivative
Placement of $C^{1,1}$	Refines the gap between C^1 and C^2
Bounded f''	Bounded second derivative gives $C^{1,1}$
Example: differentiable not C^1	Oscillatory derivative discontinuity at 0

Definition (Continuously Differentiable; Class C^1)

Definition 1.7.18 (Continuously Differentiable; Class C^1). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *continuously differentiable* on I , written $f \in C^1(I)$, if f is differentiable at every point of I and the derivative $f' : I \rightarrow \mathbb{R}$ is continuous on I .

Remark (Standard quantified statement).

$$f \in C^1(I) \iff (\forall x \in I : \text{DifferentiableAt}(f, x)) \wedge \text{ContinuousOn}(f', I).$$

Remark (Interpretation). The condition $f \in C^1(I)$ is strictly stronger than differentiability. A function may be differentiable everywhere while its derivative fails to be continuous; Darboux's Theorem still constrains that failure, because derivatives cannot jump. Thus the gap between differentiability and C^1 is oscillatory rather than jump-like.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Definition: Continuity at a Point](#)

Example (Differentiable Everywhere, but Not C^1)

Remark (Worked example). Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At the origin,

$$f'(0) = \lim_{x \rightarrow 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \rightarrow 0} x \sin(1/x) = 0$$

by squeezing. For $x \neq 0$,

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

The first term tends to 0, but $\cos(1/x)$ oscillates, so f' is not continuous at 0. Thus f is differentiable on all of $\mathbb{R}$, but $f \notin C^1(\mathbb{R})$. The failure is oscillatory, not a jump, as Darboux's theorem requires.

Definition (Class C^k)

Definition 1.7.19 (Class C^k). Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$. The function f is of *class C^k* on I , written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \dots, f^{(k)}$ exist on I and $f^{(k)}$ is continuous on I . By convention, $C^0(I)$ is the class of continuous functions on I .

Remark (Standard quantified statement).

$$f \in C^k(I) \iff (\forall j \in \{0, 1, \dots, k\} : f^{(j)} \text{ exists on } I) \wedge \text{ContinuousOn}(f^{(k)}, I).$$

Remark (Predicate reading).

$$\text{ClassC}(f, I, k).$$

For a fixed interval I , the classes are nested:

$$f \in C^{k+1}(I) \implies f \in C^k(I).$$

Remark (Negated quantified statement).

$$f \notin C^k(I) \iff \exists x \in I : [f^{(k)} \text{ fails to exist at } x] \vee [f^{(k)} \text{ exists on } I \wedge \neg \text{ContinuousAt}(f^{(k)}, x)].$$

Remark (Negation predicate reading). $\neg \text{ClassC}(f, I, k)$: somewhere the k -th derivative either does not exist or exists but is not continuous. A single point where the top derivative is discontinuous certifies $f \notin C^k(I)$.

Remark (Interpretation). The notation $C^k(I)$ records k layers of differentiability together with continuity at the top layer. This is the regularity scale used to state how many times a theorem may differentiate a function without leaving the controlled setting.

Remark (Dependencies).

- [Definition: Higher Derivatives](#)
- [Definition: Continuously Differentiable; Class \$C^1\$](#)

Definition (Smooth Function; Class C^∞)

Definition 1.7.20 (Smooth Function; Class C^∞). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *smooth*, or of *class* C^∞ , written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

Remark (Standard quantified statement).

$$f \in C^\infty(I) \iff \forall k \in \mathbb{N} : f \in C^k(I).$$

Remark (Negated quantified statement).

$$f \notin C^\infty(I) \iff \exists k \in \mathbb{N} : f \notin C^k(I).$$

A single order at which differentiability or continuity breaks demotes the function out of C^∞ .

Remark (Interpretation). Smoothness is the regularity language imported by differential geometry. Smooth manifolds are assembled from C^∞ transition maps, and the tangent-map construction seeded by the differential uses this vocabulary.

Remark (Dependencies).

- [Definition: Class \$C^k\$](#)

Definition (Real-Analytic Function; Class C^ω)

Definition 1.7.21 (Real-Analytic Function; Class C^ω). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *real-analytic*, or of *class C^ω* , written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad \text{for every } x \in (a-r, a+r) \cap I.$$

Remark (Standard quantified statement).

$$f \in C^\omega(I) \iff \forall a \in I \exists r > 0 \forall x \in (a-r, a+r) \cap I : f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Remark (Interpretation). Analytic regularity means that the Taylor series does more than approximate the function: near each point, the series equals the function. The infinite Taylor series and its convergence are developed in [Taylor's construction and error analysis](#), which supplies the series machinery behind this definition.

Remark (Dependencies).

- [Definition: Smooth Function; Class \$C^\infty\$](#)
- [Definition: Taylor Polynomial at a Point](#)

Theorem (The Smoothness Tower)

Theorem 1.7.22 (The Smoothness Tower). *For every nondegenerate interval I , the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

hold, and each displayed inclusion is strict. Go to proof.

Remark (Standard quantified statement).

$$\forall k \in \mathbb{N} : C^{k+1}(I) \subsetneq C^k(I), \quad C^\omega(I) \subsetneq C^\infty(I).$$

Remark (Failure modes). The inclusions do not fail; only their strictness needs witnesses.

- $C^k \setminus C^{k+1}$: $\varphi_k(x) = x^k|x|$, whose k -th derivative is $(k+1)!|x|$, continuous but not differentiable at 0.
- $C^\infty \setminus C^\omega$: the flat function e^{-1/x^2} , smooth with all derivatives zero at 0, so its Maclaurin series is zero while the function is not.

These separators prove every displayed containment is a genuine drop.

Remark (Interpretation). The inclusions are definitional; strictness is witnessed by examples. On any nondegenerate interval, translated and rescaled copies of $x \mapsto x^k|x|$ separate C^k from C^{k+1} . Translated copies of the flat function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

separate C^∞ from C^ω : all derivatives at the flat point vanish, so the Taylor series is identically zero, while the function is positive on one side of that point.

Remark (Dependencies).

- Definition: Class C^k
- Definition: Smooth Function; Class C^∞
- Definition: Real-Analytic Function; Class C^ω
- Theorem: Darboux's Theorem

Remark (Section guide). Between C^1 and C^2 there is a regularity floor worth naming. A C^1 function has a continuous derivative; a C^2 function has a derivative that is itself differentiable. Many functions used in analysis live in the gap: their first derivative is continuous and controlled, but a classical second derivative may fail to exist at a few points. The control is a Lipschitz bound on f' , and the class is written $C^{1,1}$.

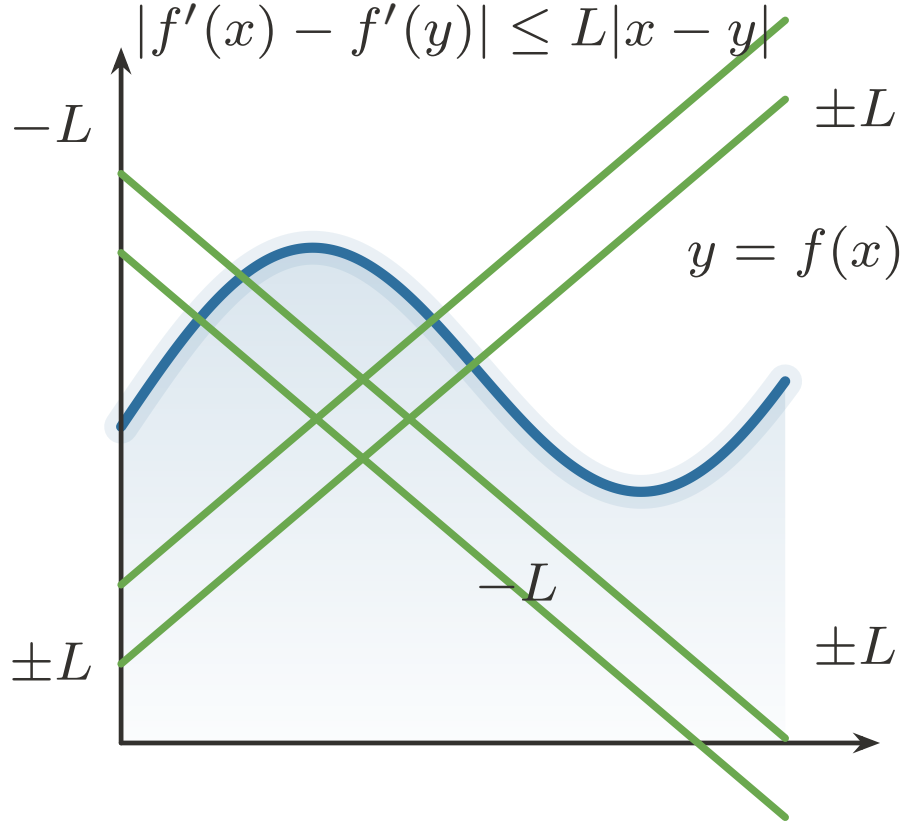


Figure 1.6: A Lipschitz derivative changes at a uniformly bounded rate.

Definition (Lipschitz-Derivative Class $C^{1,1}$)

Definition 1.7.23 (Lipschitz-Derivative Class $C^{1,1}$). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is of class $C^{1,1}$ on I , written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and f' is Lipschitz on I : there exists $L \geq 0$ such that

$$|f'(x) - f'(y)| \leq L|x - y| \quad \text{for all } x, y \in I.$$

Remark (Standard quantified statement).

$$f \in C^{1,1}(I) \iff f \in C^1(I) \wedge \exists L \geq 0 \forall x, y \in I : |f'(x) - f'(y)| \leq L|x - y|.$$

Remark (Definition predicate reading).

$$\text{ClassC11}(f, I) \iff \text{ClassC}(f, I, 1) \wedge \exists L \geq 0 : \text{Lipschitz}(f', I, L).$$

The smallest admissible L is the Lipschitz constant of f' :

$$\sup_{x \neq y} \frac{|f'(x) - f'(y)|}{|x - y|}.$$

Remark (Negated quantified statement).

$$f \notin C^{1,1}(I) \iff f \notin C^1(I) \vee (\forall L \geq 0 \exists x, y \in I : |f'(x) - f'(y)| > L|x - y|).$$

Remark (Interpretation). $C^{1,1}$ says that f' behaves as if $|f''| \leq L$, without requiring f'' to exist everywhere. It is the exact regularity tracked by many first-order estimates: the derivative is not merely continuous, but changes at a bounded rate.

Remark (Dependencies).

- Definition: Continuously Differentiable; Class C^1
- Corollary: Derivative Bound Implies Lipschitz

Theorem (Placement of $C^{1,1}$)

Theorem 1.7.24 (Placement of $C^{1,1}$). *For every nondegenerate interval I ,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

and $C^{1,1}(I)$ is not contained in $C^2(I)$. If K is a compact interval, then

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

and the first inclusion is strict. Go to proof.

Remark (Standard quantified statement).

$$(f \in C^{1,1}(I) \Rightarrow f \in C^1(I)), \quad (K \text{ compact} \wedge f \in C^2(K) \Rightarrow f \in C^{1,1}(K)),$$

with the displayed containments strict.

Remark (Failure modes). The global inclusion $C^2(I) \subseteq C^{1,1}(I)$ is false on arbitrary intervals: $f(x) = x^3$ is $C^2(\mathbb{R})$, but $f'(x) = 3x^2$ is not Lipschitz on $\mathbb{R}$. Strictness is witnessed by two elementary functions.

- $C^{1,1} \setminus C^2$: $f(x) = x|x|$ has $f'(x) = 2|x|$, which is Lipschitz, but $f''(0)$ does not exist.
- $C^1 \setminus C^{1,1}$: $f(x) = |x|^{4/3}$ has continuous first derivative, but f' has unbounded difference quotients at 0.

Remark (Interpretation). $C^{1,1}$ is a genuine intermediate regularity condition, but it is not the same as C^2 . A bounded second derivative forces the class; a Lipschitz first derivative may have corners; and a continuous first derivative may still vary too sharply to be Lipschitz.

Remark (Dependencies).

- Definition: Lipschitz-Derivative Class $C^{1,1}$
- Theorem: The Smoothness Tower
- Corollary: Derivative Bound Implies Lipschitz

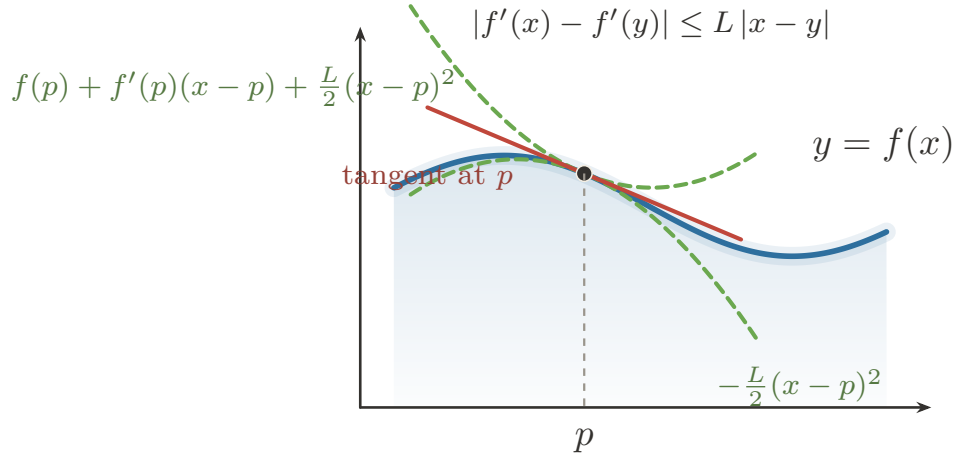


Figure 1.7: $C^{1,1}$ control gives quadratic upper and lower bounds around a tangent line.

Corollary (Bounded Second Derivative Implies $C^{1,1}$)

Corollary 1.7.25 (Bounded Second Derivative Implies $C^{1,1}$). *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \leq M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and f' is Lipschitz with constant M . Go to proof.*

Remark (Standard quantified statement).

$$(f \in C^2(I) \wedge \forall x \in I : |f''(x)| \leq M) \Rightarrow f \in C^{1,1}(I) \wedge \text{Lipschitz}(f', I, M).$$

Remark (Interpretation). This is [Derivative Bound Implies Lipschitz](#) applied one level up, to f' in place of f . The converse requires almost-everywhere language and is therefore deferred.

Remark (Dependencies).

- [Corollary: Derivative Bound Implies Lipschitz](#)
- [Definition: Class \$C^k\$](#)
- [Definition: Lipschitz-Derivative Class \$C^{1,1}\$](#)

Remark (Forward flag). A Lipschitz derivative is differentiable at most points, but making that sentence precise requires measure zero. In one variable this passes through bounded variation; in several variables it becomes Rademacher's theorem. Those results belong to the integration and measure-theoretic development, not to this chapter.

1.8 The Algebra of Derivatives

Algebra of Derivatives — Quick Reference

Rule	Role
Constant multiple	Scalars pass through differentiation
Sum	Derivative distributes over addition
Product	Differentiates pointwise products
Quotient	Differentiates quotients where denominator is nonzero
Finite sum	Iterates the sum rule over finitely many terms
Extended product	Iterates the product rule over finitely many factors
Integer power	Differentiates powers built from products
Finite linear combination	Packages linearity in finite form
Inverse Function Theorem	Produces a C^1 local inverse
Inverse derivative	Gives the reciprocal derivative formula
L'Hôpital 0/0	Evaluates indeterminate zero ratios
L'Hôpital ∞/∞	Evaluates indeterminate divergent ratios

1.8.1 From Characterization to Computation

Remark (Section guide). With Carathéodory in hand, computing derivatives becomes book-keeping with continuous slope factors. The basic rules record one structural fact: differentiation is a linear operator that also obeys a product law.

Remark (Interpretation). Carathéodory's characterization turns differentiability into continuity of a slope factor. If

$$f(x) - f(c) = (x - c)\phi_f(x), \quad g(x) - g(c) = (x - c)\phi_g(x),$$

with ϕ_f and ϕ_g continuous at c , then the standard algebraic rules for derivatives follow from the algebra of continuous functions. The recurring move is no longer to manipulate a punctured quotient directly, but to construct a continuous factor whose value at c is the derivative.

Remark (Dependencies).

- [Carathéodory Characterization of Differentiability](#)
- [Algebra of Limits](#)

Theorem (Constant Multiple Rule)

Theorem 1.8.1 (Constant Multiple Rule). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let $c \in A$ be a limit point of A , and let $\alpha \in \mathbb{R}$. If f is differentiable at c , then αf is differentiable at c , and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \text{DerivativeAt}(\alpha f, c, \alpha f'(c)).$$

Remark (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar α factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Constant Multiple Rule for Limits](#)

Theorem (Sum Rule)

Theorem 1.8.2 (Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If f and g are differentiable at c , then $f + g$ is differentiable at c , and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(f + g, c, f'(c) + g'(c)).$$

Remark (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Sum Rule for Limits](#)

Theorem (Product Rule)

Theorem 1.8.3 (Product Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If f and g are differentiable at c , then fg is differentiable at c , and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(fg, c, f'(c)g(c) + f(c)g'(c)).$$

Remark (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of f weighted by the current value of g , plus the rate of change of g weighted by the current value of f . The proof inserts the auxiliary term $f(x)g(c)$ to decouple the increments of f and g , isolating the individual difference quotients. Continuity of f at c (which follows from differentiability) is used to evaluate $\lim_{x \rightarrow c} f(x) = f(c)$ in one of the factors.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Theorem: Differentiable Implies Continuous](#)
- [Sum Rule for Limits](#)
- [Product Rule for Limits](#)

Theorem (Quotient Rule)

Theorem 1.8.4 (Quotient Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c , and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \wedge g(c) \neq 0 \Rightarrow \text{DerivativeAt}\left(\frac{f}{g}, c, \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}\right).$$

Remark (Interpretation). The quotient rule is the product rule applied to $f \cdot g^{-1}$, but stated directly. The hypothesis $g(c) \neq 0$ ensures the quotient is defined near c — continuity of g (from its differentiability) and the quotient rule for limits guarantee $g(x) \neq 0$ in a deleted neighbourhood of c , so the difference quotient of f/g is well-defined throughout. The add-and-subtract technique with $f(c)g(c)$ decouples the increments of f and g in the numerator, exactly as in the product rule proof.

Remark (Dependencies).

- [Definition: Derivative at a Point](#)
- [Theorem: Differentiable Implies Continuous](#)
- [Sum Rule for Limits](#)
- [Product Rule for Limits](#)
- [Quotient Rule for Limits](#)

1.8.2 Closure

Corollary

Corollary 1.8.5 (Finite Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \rightarrow \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A . If each f_i is differentiable at c , then $\sum_{i=1}^n \alpha_i f_i$ is differentiable at c , and*

$$\left(\sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \in \{1, \dots, n\} \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\sum_i \alpha_i f_i, c, \sum_i \alpha_i f'_i(c)\right).$$

Remark (Interpretation). Finite sums and scalar weights do not introduce a new kind of differentiation rule; they package repeated use of the Sum Rule and Constant Multiple Rule.

Remark (Dependencies).

- [Theorem: Sum Rule](#)
- [Theorem: Constant Multiple Rule](#)

Corollary

Corollary 1.8.6 (Extended Product Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If each f_i is differentiable at c , then $\prod_{i=1}^n f_i$ is differentiable at c , and*

$$\left(\prod_{i=1}^n f_i \right)'(c) = \sum_{k=1}^n f'_k(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\prod_i f_i, c, \sum_k f'_k(c) \prod_{i \neq k} f_i(c)\right).$$

Remark (Interpretation). In a finite product, each term takes one turn being differentiated while all other factors are held at their values.

Remark (Dependencies).

- [Theorem: Product Rule](#)

Corollary

Corollary 1.8.7 (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let $c \in A$ be a limit point of A , and let $n \in \mathbb{N}$. If f is differentiable at c , then f^n is differentiable at c , and*

$$(f^n)'(c) = n(f(c))^{n-1}f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge n \in \mathbb{N} \Rightarrow \text{DerivativeAt}(f^n, c, n(f(c))^{n-1}f'(c)).$$

Remark (Interpretation). The integer power rule is the extended product rule applied to n identical factors.

Remark (Dependencies).

- [Corollary: Extended Product Rule](#)

Theorem

Theorem 1.8.8 (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \rightarrow \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A . If each f_i is differentiable at c , then*

$$\left(\sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}(\sum_i \alpha_i f_i, c, \sum_i \alpha_i f'_i(c)).$$

Remark (Interpretation). The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules — it is their inductive closure. Stated in operator language: if $\mathcal{D}(I)$ denotes the vector space of functions differentiable at c , then $D_c : \mathcal{D}(I) \rightarrow \mathbb{R}$ defined by $D_c(f) = f'(c)$ is a linear functional. The Finite Linear Combination Rule says $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$.

Remark (Dependencies).

- [Real-Linear Rule](#)
- [Linear Combination of Real-Valued Functions](#)
- [Theorem: Constant Multiple Rule](#)
- [Theorem: Sum Rule](#)

1.8.3 Global Forms on an Interval

Theorem

Theorem 1.8.9 (Constant Multiple Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I , then αf is differentiable on I , and*

$$(\alpha f)'(x) = \alpha f'(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \text{ DifferentiableAt}(f, x) \Rightarrow \forall x \in I \text{ DerivativeAt}(\alpha f, x, \alpha f'(x)).$$

Remark (Interpretation). The pointwise constant-multiple rule becomes an interval statement by applying it at every point of the interval.

Remark (Dependencies).

- [Theorem: Constant Multiple Rule](#)

Theorem

Theorem 1.8.10 (Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \rightarrow \mathbb{R}$. If f and g are differentiable on I , then $f + g$ is differentiable on I , and*

$$(f + g)'(x) = f'(x) + g'(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I (\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x)) \Rightarrow \forall x \in I \text{ DerivativeAt}(f+g, x, f'(x)+g'(x)). \blacksquare$$

Remark (Interpretation). The derivative of a sum is computed point by point across the whole interval.

Remark (Dependencies).

- [Theorem: Sum Rule](#)

Theorem

Theorem 1.8.11 (Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \rightarrow \mathbb{R}$. If f and g are differentiable on I , then fg is differentiable on I , and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \left(\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x) \right) \Rightarrow \forall x \in I \text{ DerivativeAt}(fg, x, f'(x)g(x) + f(x)g'(x))$$

Remark (Interpretation). The interval product rule records the product rule as an identity of derivative functions.

Remark (Dependencies).

- [Theorem: Product Rule](#)

Theorem

Theorem 1.8.12 (Quotient Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \rightarrow \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I , and*

$$\left(\frac{f}{g} \right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \left(\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x) \wedge g(x) \neq 0 \right) \Rightarrow \forall x \in I \text{ DerivativeAt}\left(\frac{f}{g}, x, \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}\right)$$

Remark (Interpretation). The quotient rule on an interval is the pointwise quotient rule with the nonvanishing denominator hypothesis required at every point.

Remark (Dependencies).

- [Theorem: Quotient Rule](#)

Corollary

Corollary 1.8.13 (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \rightarrow \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I , then $\sum_{i=1}^n \alpha_i f_i$ is differentiable on I , and*

$$\left(\sum_{i=1}^n \alpha_i f_i \right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \forall x \in I \text{ DifferentiableAt}(f_i, x) \Rightarrow \forall x \in I \text{ DerivativeAt}\left(\sum_i \alpha_i f_i, x, \sum_i \alpha_i f'_i(x)\right).$$

Remark (Interpretation). Finite linear combinations commute with differentiation as functions on the whole interval.

Remark (Dependencies).

- Corollary: Finite Sum Rule

Corollary

Corollary 1.8.14 (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \dots, f_n : I \rightarrow \mathbb{R}$. If each f_i is differentiable on I , then $\prod_{i=1}^n f_i$ is differentiable on I , and*

$$\left(\prod_{i=1}^n f_i \right)'(x) = \sum_{k=1}^n f'_k(x) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \forall x \in I \text{ DifferentiableAt}(f_i, x) \Rightarrow \forall x \in I \text{ DerivativeAt}\left(\prod_i f_i, x, \sum_k f'_k(x) \prod_{i \neq k} f_i(x)\right).$$

Remark (Interpretation). The product rule for many factors holds uniformly point by point on the interval.

Remark (Dependencies).

- Corollary: Extended Product Rule

Corollary

Corollary 1.8.15 (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I , then f^n is differentiable on I , and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \text{ DifferentiableAt}(f, x) \wedge n \in \mathbb{N} \Rightarrow \forall x \in I \text{ DerivativeAt}(f^n, x, n(f(x))^{n-1} f'(x)).$$

Remark (Interpretation). The interval power rule says that the pointwise integer-power formula assembles into an equality of derivative functions.

Remark (Dependencies).

- Corollary: Power Rule for Integer Powers of a Function

Theorem

Theorem 1.8.16 (Finite Linear Combination Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \rightarrow \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I , then*

$$\left(\sum_{i=1}^n \alpha_i f_i \right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \forall x \in I \text{ DifferentiableAt}(f_i, x) \Rightarrow \forall x \in I \text{ DerivativeAt}(\sum_i \alpha_i f_i, x, \sum_i \alpha_i f'_i(x)).$$

Remark (Interpretation). The pointwise statements above — each proved at a single point c — extend to global statements when the hypotheses hold at every point of I . If f is differentiable on I , then $f' : I \rightarrow \mathbb{R}$ is itself a function and differentiation acts as a linear operator on the vector space $\mathcal{D}(I)$ of functions differentiable on I :

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g} \right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on I . The Finite Linear Combination Rule in its global form is precisely the statement that $D : \mathcal{D}(I) \rightarrow \mathcal{F}(I)$ defined by $D(f) = f'$ is a linear map between function spaces, where $\mathcal{F}(I)$ denotes functions on I . The interval variants are not separate theorems — they are the pointwise results collected into function notation by applying the pointwise result at each $x \in I$.

Remark (Dependencies).

- [Real-Linear Rule](#)
- [Linear Combination of Real-Valued Functions](#)
- [Theorem: Finite Linear Combination Rule](#)

When f is strictly monotone and differentiable on I with $f'(x) \neq 0$ for all $x \in I$, the inverse $g : J \rightarrow I$ is differentiable on $J = f(I)$ and

$$g' = \frac{1}{f' \circ g},$$

as functions on J . This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

1.8.4 Inverting the Operation

Theorem (Inverse Function Theorem, One Variable)

Theorem 1.8.17 (Inverse Function Theorem, One Variable). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$ be of class C^1 , and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. $f|_U : U \rightarrow V$ is a bijection;
2. the inverse $g = (f|_U)^{-1} : V \rightarrow U$ is of class C^1 ;
3. for every $y \in V$,

$$g'(y) = \frac{1}{f'(g(y))}.$$

Go to proof.

Remark (Standard quantified statement).

$$f \in C^1(I) \wedge c \in I \wedge f'(c) \neq 0$$

$$\implies \exists U \ni c \exists V \ni f(c) [\text{Bijective}(f|_U, U, V) \wedge (f|_U)^{-1} \in C^1(V) \wedge \forall y \in V : ((f|_U)^{-1})'(y) = 1/f'((f|_U)^{-1}(y))]$$

Remark (Predicate reading). $\text{LocallyInvertible}(f, c)$ holds with a C^1 local inverse. The condition $f'(c) \neq 0$ upgrades local set-theoretic invertibility to local analytic invertibility: the inverse exists as a differentiable function and has controlled derivative.

Remark (Failure modes). The hypotheses fail in different ways.

- If $f'(c) = 0$, the inverse may exist without being differentiable. The map $f(x) = x^3$ is globally bijective, but its inverse $x \mapsto x^{1/3}$ is not differentiable at 0.
- If f' is not continuous at c , the sign of f' need not persist near c . Mere differentiability with $f'(c) \neq 0$ is therefore not enough to force local monotonicity or local invertibility.

Remark (Contrapositive quantified statement).

$$\neg \text{LocallyInvertible}_{C^1}(f, c) \implies f \notin C^1(\text{near } c) \vee f'(c) = 0.$$

Remark (Contrapositive predicate reading). If f has no C^1 local inverse at c , then either f fails to be continuously differentiable near c , or its derivative vanishes there. The witnesses $x \mapsto x^3$ and $x \mapsto x + 2x^2 \sin(1/x)$ isolate the two failure mechanisms.

Remark (Interpretation). This theorem is the existence result behind the inverse-derivative formula. A continuous nonzero derivative keeps its sign on a small interval; constant sign forces strict monotonicity; strict monotonicity gives a local inverse; and the Carathéodory characterization upgrades that inverse to a differentiable map with reciprocal slope. In several variables, the condition $f'(c) \neq 0$ becomes invertibility of the differential.

Remark (Dependencies).

- Definition: Continuously Differentiable; Class C^1
- Theorem: Monotonicity from the Sign of the Derivative
- Theorem: Continuous Inverse Theorem
- Theorem: Carathéodory Characterization of Differentiability

Corollary (Derivative of the Inverse Function)

Corollary 1.8.18 (Derivative of the Inverse Function). *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse g satisfies*

$$g' = \frac{1}{f' \circ g}$$

on V . More globally, if $f : I \rightarrow J$ is a C^1 bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \rightarrow I$ is C^1 and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \quad \text{for every } y \in J.$$

Go to proof.

Remark (Standard quantified statement).

$$g = (f|_U)^{-1} \implies \forall y \in V : g'(y) = \frac{1}{f'(g(y))}.$$

Remark (Interpretation). The formula says that undoing a differentiable change of variable reverses the local stretch factor. If f multiplies small increments near x by $f'(x)$, then its inverse multiplies corresponding increments near $f(x)$ by $1/f'(x)$.

Remark (Dependencies).

- Theorem: Inverse Function Theorem, One Variable
- Theorem: Chain Rule

1.8.5 Indeterminate Forms: L'Hôpital's Rule

Theorem (L'Hôpital's Rule, 0/0 Form)

Theorem 1.8.19 (L'Hôpital's Rule, 0/0 Form). *Let $a < b$, and let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

Remark (Standard quantified statement).

$$\left(\lim_{x \rightarrow a^+} f(x) = 0 \wedge \lim_{x \rightarrow a^+} g(x) = 0 \wedge \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \right) \implies \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

Remark (Failure modes). The rule is a one-way implication and is routinely misapplied.

- *Form not indeterminate.* If f/g is not genuinely 0/0, differentiating numerator and denominator can answer a different problem.
- *$\lim f'/g'$ fails to exist.* With $f(x) = x + \sin x$ and $g(x) = x$ as $x \rightarrow +\infty$, the quotient $f/g \rightarrow 1$, but $f'/g' = 1 + \cos x$ oscillates. Absence of $\lim f'/g'$ says nothing by itself about $\lim f/g$.
- *g' vanishes near the target.* The denominator derivative must stay nonzero so the Cauchy Mean Value Theorem applies and the derivative ratio is defined.

Remark (Interpretation). The 0/0 form is the Cauchy Mean Value Theorem read as a limit law. On a shrinking interval, a ratio of changes in f and g is represented by a ratio of derivatives at an interior point. If the derivative ratio settles, the original quotient is forced to settle to the same value.

Remark (Dependencies).

- [Theorem: Cauchy Mean Value Theorem](#)
- [Definition: Limit of a Function](#)
- [Theorem: Squeeze Theorem for Function Limits](#)

Theorem (L'Hôpital's Rule, ∞/∞ Form)

Theorem 1.8.20 (L'Hôpital's Rule, ∞/∞ Form). *Let $a < b$, and let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \rightarrow a^+} |g(x)| = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on $\lim_{x \rightarrow a^+} f(x)$ is required. Go to proof.

Remark (Standard quantified statement).

$$\left(\lim_{x \rightarrow a^+} |g(x)| = +\infty \wedge \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \right) \implies \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

Remark (Failure modes). L'Hôpital's Rule is a one-way implication and requires an indeterminate form. If the quotient is not of type $0/0$ or ∞/∞ , differentiating numerator and denominator can change the problem. The derivative ratio must also have a limit: for $f(x) = x + \sin x$ and $g(x) = x$ as $x \rightarrow +\infty$, the quotient f/g tends to 1, but $f'/g' = 1 + \cos x$ has no limit.

Remark (Interpretation). The ∞/∞ form compares two functions whose magnitudes escape. The Cauchy Mean Value Theorem controls the quotient after subtracting fixed endpoint values, and the divergent denominator makes those fixed values negligible.

Remark (Dependencies).

- [Theorem: Cauchy Mean Value Theorem](#)
- [Definition: Limit of a Function](#)
- [Theorem: Squeeze Theorem for Function Limits](#)

1.9 Taylor: Construction and Its Error

Taylor Expansion — Quick Reference

Concept	Meaning
Taylor polynomial	Polynomial forced by derivative data at a point
Taylor remainder	Exact error after subtracting the Taylor polynomial
Maclaurin polynomial	Taylor polynomial centered at 0
Lagrange remainder	Error controlled by an intermediate derivative value
Peano remainder	Local asymptotic error smaller than the retained order
Example: $(\sin x - x)/x^3$	Taylor and L'Hôpital compared on one limit
Flat function	Smooth function whose Taylor series misses it

Taylor's theorem is the derivative's most ambitious claim: local behaviour to finite order is encoded by derivatives at one point, with an error term that can be estimated. The first-order case is the seed of the closing reframe, where the derivative stops being only a number $f'(c)$ and becomes the linear map $h \mapsto f'(c)h$.

Definition (Taylor Polynomial at a Point)

Definition 1.9.1 (Taylor Polynomial at a Point). Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \leq k \leq n$. The **Taylor polynomial of order n** for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$, $a \in I$, $n \in \mathbb{N}$, $f : I \rightarrow \mathbb{R}$.

If $f^{(k)}(a)$ exists for every $0 \leq k \leq n$, then

$$\forall x \in I \left(T_{n,a}f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k \right).$$

Remark (Interpretation). The Taylor polynomial is the unique polynomial of degree at most n whose first n derivatives at a agree with those of f . Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

Remark (Dependencies).

- [Derivative at a Point](#)
- [Power Rule for Integer Powers of a Function](#)
- [Finite Sum Rule](#)

Definition (Taylor Remainder)

Definition 1.9.2 (Taylor Remainder). Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order n** at a is the function $R_{n,a}f : I \rightarrow \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Remark (Standard quantified statement).

$$\forall x \in I \left(R_{n,a}f(x) = f(x) - T_{n,a}f(x) \right).$$

Remark (Interpretation). The remainder is the exact error made by replacing f with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

Remark (Dependencies).

- Taylor Polynomial at a Point

Definition (Maclaurin Polynomial)

Definition 1.9.3 (Maclaurin Polynomial). Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \leq k \leq n$. The **Maclaurin polynomial of order n** for f is

$$T_{n,0}f(x) := \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k.$$

Remark (Standard quantified statement).

$$\forall x \in I \left(T_{n,0}f(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k \right).$$

Remark (Interpretation). Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

Remark (Dependencies).

- Taylor Polynomial at a Point

Theorem (Taylor's Theorem with Lagrange Remainder)

Theorem 1.9.4 (Taylor's Theorem with Lagrange Remainder). Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \rightarrow \mathbb{R}$. Suppose that $f, f', \dots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b) . Then, for every $x \in (a, b)$, there exists c strictly between a and x such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b) \exists c \in (a, x) \left(f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right).$$

Remark (Interpretation). Taylor's theorem says that the error after truncating at degree n has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point c is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Mean Value Theorem
- Finite Sum Rule
- Power Rule for Integer Powers of a Function

Corollary (Taylor Expansion with Peano Remainder)

Corollary 1.9.5 (Taylor Expansion with Peano Remainder). *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$ have derivatives up to order n in a neighbourhood of a . If $f^{(n)}$ is continuous at a , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + o((x-a)^n) \quad \text{as } x \rightarrow a.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x \rightarrow a} \frac{f(x) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}{(x-a)^n} = 0.$$

Remark (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order n , and the remaining error is smaller than $(x-a)^n$ near the center.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Limit of a Function
- Derivative at a Point

Corollary (First-Order Peano Remainder)

Corollary 1.9.6 (First-Order Peano Remainder). *Let f be differentiable at c . Then, as $h \rightarrow 0$,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \lim_{h \rightarrow 0} \frac{f(c+h) - f(c) - f'(c)h}{h} = 0.$$

Remark (Interpretation). The first-order Peano formula is the derivative rewritten as an approximation theorem. It says that the affine approximation $f(c) + f'(c)h$ captures all first-order change, and the remaining error is smaller than h itself.

Remark (Dependencies).

- [Derivative at a Point](#)

Example (($\sin x - x$)/ x^3 Two Ways)

Remark (Worked example). Taylor expansion gives

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$

so

$$\frac{\sin x - x}{x^3} = \frac{-x^3/6 + o(x^3)}{x^3} \rightarrow -\frac{1}{6}.$$

L'Hôpital's rule reaches the same value by three differentiations:

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{6x} = \lim_{x \rightarrow 0} \frac{-\cos x}{6} = -\frac{1}{6}.$$

Taylor exposes the reason for the answer: the limit is the cubic coefficient.

Proposition (The Flat Function: Smooth but Not Analytic)

Proposition 1.9.7 (The Flat Function: Smooth but Not Analytic). *Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then:

1. $f \in C^\infty(\mathbb{R})$;
2. $f^{(n)}(0) = 0$ for every $n \geq 0$;
3. the Maclaurin series of f is identically 0, so $f \notin C^\omega$ on any neighbourhood of 0.

In particular, $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \emptyset$. Go to proof.

Remark (Standard quantified statement).

$$(\forall n \in \mathbb{N} : f^{(n)}(0) = 0) \wedge (\forall r > 0 \exists x : 0 < |x| < r \wedge f(x) \neq 0) \implies f \in C^\infty \setminus C^\omega.$$

Remark (Interpretation). The Taylor series sees only derivative data at the center, and all of that data is zero for the flat function at 0. The function is nevertheless positive away from 0. This separates smoothness from analyticity and supplies the strictness witness referenced by the smoothness tower.

Remark (Dependencies).

- Definition: Higher Derivatives
- Definition: Smooth Function; Class C^∞
- Definition: Real-Analytic Function; Class C^ω
- Definition: Maclaurin Polynomial
- Theorem: The Smoothness Tower

1.9.1 The Differential

Remark (Interpretation). The derivative value $f'(c) \in \mathbb{R}$ is a number. The differential df_c is a linear map from increments to first-order changes. In one dimension these two objects are represented by the same scalar, but they play different roles.

Definition (Differentiability by a Differential)

Definition 1.9.8 (Differentiability by a Differential). Let f be defined on a neighbourhood of $c \in \mathbb{R}$. The function f is *differentiable at c in the differential sense* if there exists a linear map $L : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(c + h) - f(c) = L(h) + o(h) \quad \text{as } h \rightarrow 0.$$

In that case the *differential* of f at c is $df_c := L$.

Remark (Standard quantified statement).

$$\exists L : \mathbb{R} \rightarrow \mathbb{R} \text{ linear such that } f(c + h) - f(c) = L(h) + o(h).$$

Remark (Interpretation). The differential is the linear part of the increment. The function itself is approximated affinely by $f(c) + df_c(x - c)$, while df_c alone is the linear map acting on the displacement.

Remark (Dependencies).

- First-Order Peano Remainder

Theorem (Differential and Derivative Agree)

Theorem 1.9.9 (Differential and Derivative Agree). *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. Then f is differentiable at c in the ordinary derivative sense if and only if it is differentiable at c in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \exists df_c \text{ with } df_c(h) = f'(c)h.$$

Remark (Interpretation). This theorem is the bridge from the slope number to the linear map. The one-dimensional derivative is multiplication by the scalar $f'(c)$.

Remark (Dependencies).

- [Derivative at a Point](#)
- [Differentiability by a Differential](#)

Theorem (Uniqueness of the Differential)

Theorem 1.9.10 (Uniqueness of the Differential). *If $L_1, L_2 : \mathbb{R} \rightarrow \mathbb{R}$ are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \quad \text{as } h \rightarrow 0,$$

then $L_1 = L_2$. Go to proof.

Remark (Standard quantified statement).

$$(L_1(h) - L_2(h) = o(h)) \implies L_1 = L_2.$$

Remark (Interpretation). The differential is not a choice of linear approximation among many. If a first-order linear part exists, it is forced.

Remark (Dependencies).

- [Differentiability by a Differential](#)

Theorem (Differential Continuity Criterion)

Theorem 1.9.11 (Differential Continuity Criterion). *If f is differentiable at c in the differential sense, then f is continuous at c . Go to proof.*

Remark (Standard quantified statement).

$$\text{DifferentiableByDifferentialAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

Remark (Interpretation). From $f(c+h) - f(c) = df_c(h) + o(h)$, both terms on the right tend to zero as $h \rightarrow 0$. Continuity is therefore immediate in the differential language.

Remark (Dependencies).

- [Differentiability by a Differential](#)

Theorem (Chain Rule for Differentials)

Theorem 1.9.12 (Chain Rule for Differentials). *If f is differentiable at c and g is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Go to proof.

Remark (Standard quantified statement).

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Remark (Interpretation). In one dimension this says that composing two scalar multiplications multiplies their scalars. In several variables the same formula becomes matrix multiplication.

Remark (Dependencies).

- [Chain Rule](#)
- [Differential and Derivative Agree](#)

Theorem (Linearity of the Differential)

Theorem 1.9.13 (Linearity of the Differential). *If f and g are differentiable at c and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c.$$

Go to proof.

Remark (Standard quantified statement).

$$d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c.$$

Remark (Interpretation). The differential packages the linearity of differentiation as equality of linear maps.

Remark (Interpretation). On a manifold, the same construction becomes a tangent map

$$T_c f : T_c M \rightarrow T_{f(c)} N.$$

The one-dimensional differential is the prototype: replace real increments by tangent vectors and replace scalar multiplication by a linear map between tangent spaces.

Remark (Dependencies).

- [Finite Linear Combination Rule](#)
- [Differential and Derivative Agree](#)

Bibliography